

Chapter 7

Epidemiology of Periodontal Diseases

Panos N. Papapanou¹ and Jan Lindhe²

¹Division of Periodontics, Section of Oral and Diagnostic Sciences,
Columbia University College of Dental Medicine, New York, NY, USA

²Department of Periodontology, Institute of Odontology,
The Sahlgrenska Academy at University of Gothenburg, Gothenburg, Sweden

Introduction, 125	Periodontitis and tooth loss, 138
Methodologic issues, 125	Risk factors for periodontitis, 138
Examination methods: Index systems, 125	Introduction: Definitions, 138
Periodontitis “case definition” in epidemiologic studies, 127	Non-modifiable background factors, 141
Prevalence of periodontal diseases, 130	Environmental, acquired, and behavioral factors, 143
Periodontitis in adults, 130	Concluding remarks, 154
Periodontitis in children and adolescents, 135	Acknowledgments, 156

Introduction

The term epidemiology is of Hellenic origin; it consists of the preposition “epi”, which means “among” or “against”, and the noun “demos”, which means “people”. As denoted by its etymology, epidemiology is defined as “the study of the distribution of disease or a physiological condition in human populations and of the factors that influence this distribution” (Lilienfeld 1978). A more inclusive description by Frost (1941) emphasizes that “epidemiology is essentially an inductive science, concerned not merely with describing the distribution of disease, but equally or more with fitting it into a consistent philosophy”. Thus, the information obtained from an epidemiologic investigation should extend beyond a mere description of the distribution of the disease in different populations (*descriptive* epidemiology). It should be further expanded to (1) elucidate the etiology of a specific disease by combining epidemiologic data with information from other disciplines such as genetics, biochemistry, microbiology, sociology, etc. (*etiological* epidemiology); (2) evaluate the consistency of epidemiologic data with hypotheses developed clinically or experimentally (*analytical* epidemiology); and (3) provide the basis for developing and evaluating preventive procedures and public health practices (*experimental* epidemiology).

Based on the above, epidemiologic research in periodontology must (1) fulfil the task of providing data on the *prevalence* of periodontal diseases (the frequency of their occurrence) in different populations, as well as on the *severity* of such conditions, that is the amount of pathologic changes; (2) elucidate aspects related to the *etiology* and the *determinants* of these diseases (*causative* and *risk* factors); and (3) provide documentation concerning the effectiveness of preventive and therapeutic measures on a population basis.

Methodologic issues

Examination methods: Index systems

Examination of the periodontal status of a given individual includes clinical assessments of inflammation in the gingiva, recordings of probing depths and clinical attachment levels, as well as radiographic assessments of the amount of loss of supporting alveolar bone. A variety of index systems for the scoring of these parameters have been developed, some of which were designed exclusively for examination of patients in a dental practice set-up, while others were developed for use in epidemiologic research. The design of the index systems and the definition

of the various scores inevitably reflect the knowledge of the etiology and pathogenesis of periodontal diseases at the time these systems were introduced, as well as concepts related to the contemporary therapeutic approaches and strategies. This section will not provide a complete list of all available scoring systems, but rather give a brief description of a limited number of indices that are either currently used or are likely to be encountered in the recent literature. For a description of earlier scoring systems and a historical perspective of their development, the reader is referred to Ainamo (1989).

Assessment of inflammation of the periodontal tissues

Presence of inflammation in the gingiva is usually recorded by the use of a probe, and often according to the principles of the Gingival Index System outlined by Löe (1967). According to this system, absence of visual signs of inflammation in the gingival unit is scored as 0, while a slight change in color and texture is scored as 1. Visual inflammation and bleeding tendency from the gingival margin right after a periodontal probe is briefly run along the gingival margin is scored as 2, while overt inflammation with a tendency for spontaneous bleeding is scored as 3. A parallel index scores plaque deposits (Plaque Index System) on a scale from 0 to 3 (Silness & Löe 1964): the absence of plaque is scored as 0, plaque disclosed after running the periodontal probe along the gingival margin as 1, visible plaque as 2, and abundant plaque as 3. Simplified variants of both the Gingival and the Plaque indices (Ainamo & Bay 1975) have been extensively used, assessing presence/absence of inflammation or plaque, respectively, in a binomial fashion (*dichotomous scoring*). In such systems, bleeding from the gingival margin and visible plaque correspond to a score of 1, while absence of bleeding and no visible plaque correspond to a score of 0.

Bleeding after probing to the base of the probable pocket (Gingival Sulcus Bleeding Index) has been a common way of establishing the occurrence of subgingival inflammation, characterized by the presence of an inflammatory infiltrate adjacent to the ulcerated pocket epithelium (Mühlemann & Son 1971). In this dichotomous registration, cases where bleeding emerges within 15 seconds after probing are scored as 1.

Assessment of loss of periodontal tissue support

One of the early indices providing indirect information on the loss of periodontal tissue support was the Periodontal Index (PI) developed in the 1950s by Russell (1956), and until the 1980s it was the most widely used index in epidemiologic studies of periodontal disease. Its criteria are applied to each

tooth and the scoring is as follows: a tooth with healthy periodontium scores 0, a tooth with gingivitis around only part of the tooth circumference scores 1, a tooth with gingivitis encircling the tooth scores 2, pocket formation scores 6, and loss of function due to excessive tooth mobility scores 8. Due to the nature of the criteria used, the PI is a reversible scoring system; in other words, after treatment, a tooth or an individual can have the score lowered or reduced to 0.

In contrast to the PI system, the Periodontal Disease Index (PDI), developed by Ramfjord (1959), is a system designed to assess *destructive* disease; it measures *loss of attachment* instead of *pocket depth* and is, therefore, an irreversible index. The scores, ranging from 0 to 6, denote periodontal health or gingivitis (scores 0–3) and various levels of attachment loss (scores 4–6).

In contemporary epidemiologic studies, loss of periodontal tissue support is assessed by measurements of probing pocket depth (PPD) and probing attachment level (PAL). PPD is defined as the distance from the gingival margin to the apical location of the tip of a periodontal probe that is inserted into the pocket using a moderate probing force. Likewise, PAL or clinical attachment level (CAL) is defined as the distance from the cemento-enamel junction (CEJ) to the location of the probe tip. Probing assessments are usually carried out at several locations along the tooth circumference (buccal, lingual, mesial, and distal). The number of probing assessments per tooth has varied in epidemiologic studies from two to six, while the examination may include either all teeth present (*full-mouth*) or a subset of *index* teeth (*partial-mouth* examination).

Carlos *et al.* (1986) proposed an index system which records loss of periodontal tissue support. The index was denoted the Extent and Severity Index (ESI) and consists of two components (*bivariate* index): (1) the *Extent*, describing the proportion of tooth sites of a subject showing signs of destructive periodontitis, and (2) the *Severity*, describing the amount of PAL at the diseased sites, expressed as a mean value. An attachment loss threshold of >1 mm was set as the criterion that qualified a tooth site as affected by the disease. The introduction of a threshold value serves a dual purpose: (1) it readily distinguishes the fraction of the dentition affected by disease at levels exceeding the error inherent in the clinical measurement of attachment loss; and (2) it prevents unaffected tooth sites from contributing to the individual subject's mean attachment loss value. In order to limit the number of measurements to be performed, a partial examination comprising the mid-buccal and mesiobuccal aspects of the upper right and lower left quadrants was recommended. It has to be emphasized that the system was designed to assess the cumulative effect of destructive periodontal disease rather than the presence of the disease itself. The bivariate nature of the index facilitates a

rather detailed description of attachment loss patterns: for example, an ESI of (90, 2.5) suggests a generalized but rather mild form of destructive disease, in which 90% of the tooth sites are affected by an average attachment loss of 2.5 mm. In contrast, an ESI of (20, 7.0) describes a severe, localized form of disease. Validation of various partial extent and severity scoring systems against the full-mouth estimates has been performed (Papapanou *et al.* 1993).

Radiographic assessment of alveolar bone loss

The potential and the limitations of intraoral radiography to describe loss of supporting periodontal tissues were reviewed in classical publications (Lang & Hill 1977; Benn 1990) and recent reports (Vandenberghe *et al.* 2010). Radiographs have been commonly employed in cross-sectional epidemiologic studies to evaluate the result of periodontal disease on the supporting tissues rather than the presence of the disease itself, and are thought to provide valid estimates of the extent and severity of destructive periodontitis (Pitiphat *et al.* 2004). Radiographic assessments have been particularly common as screening methods for detecting subjects suffering from juvenile periodontitis as well as a means for monitoring periodontal disease progression in longitudinal studies. Assessments of bone loss in intraoral radiographs are usually performed by evaluating a multitude of qualitative and quantitative features of the visualized interproximal bone, including (1) the presence of an intact lamina dura, (2) the width of the periodontal ligament space, (3) the morphology of the bone crest ("even" or "angular" appearance), and (4) the distance between the CEJ and the most coronal level at which the periodontal ligament space is considered to exhibit normal width. The threshold for bone loss (the CEJ–bone crest distance considered to indicate that bone loss has occurred) varies between 1 and 3 mm in different studies. Radiographic data are usually presented as (1) mean bone loss scores per subject (or group of subjects) and (2) number or percentage of tooth surfaces per subject (or group of subjects) exhibiting bone loss exceeding certain thresholds. In early studies, bone loss was frequently recorded using "ruler" devices, describing the amount of lost or remaining bone as a percentage of the length of the root or the tooth (Schei *et al.* 1959; Lavstedt *et al.* 1975).

Assessment of periodontal treatment needs

An index system aimed at assessing the need for periodontal treatment in large population groups was developed, on the initiative of the World Health Organization (WHO), by Ainamo *et al.* (1982). The principles of the Community Periodontal Index for Treatment Needs (CPITN) can be summarized as follows:

1. The dentition is divided into six *sextants* (one anterior and two posterior tooth regions in each dental arch). The treatment need in a sextant is recorded when two or more teeth not intended for extraction are present. If only one tooth remains in the sextant, the tooth is included in the adjoining sextant.
2. Probing assessments are performed either around all teeth in a sextant or around certain index teeth (the latter approach has been recommended for epidemiologic surveys). Only the most severe measure in the sextant is chosen to represent the sextant.
3. The periodontal conditions are scored as follows:
 - *Code 0* is given to a sextant with no pockets, calculus or overhangs of fillings and no bleeding on probing
 - *Code 1* is given to a sextant with no pockets, calculus or overhangs of fillings, but in which bleeding occurs after gentle probing in one or several gingival units
 - *Code 2* is assigned to a sextant if there are no teeth with pockets exceeding 3 mm, but in which dental calculus and plaque-retaining factors are identified subgingivally
 - *Code 3* is given to a sextant that harbors teeth with 4–5-mm deep pockets
 - *Code 4* is given to a sextant that harbors teeth with pockets that are 6 mm deep or deeper.
4. The treatment needs (TNs) scores range from 0 to 4 and are based on the most severe periodontal condition code in the entire dentition, recorded as above. Thus, TN 0 indicates no need for periodontal therapy in the presence of gingival health (*Code 0*); TN 1 need for improved oral hygiene (*Code 1*); TN 2 need for scaling, removal of overhangs, and improved oral hygiene (*Codes 2+3*); and TN 3 more advanced treatment needs (*Code 4*).

Although not designed for epidemiologic purposes, this index system has been extensively used, and CPITN-based studies have often been the sole source of epidemiologic information on periodontal conditions, particularly those from developing countries. A later modification of the index, termed Community Periodontal Index (WHO 1997), places more emphasis on the assessment of periodontal conditions rather than the assessment of periodontal treatment needs. A substantial amount of data generated by the use of CPITN/ CPI have been accumulated in the WHO Global Oral Data Bank (Miyazaki *et al.* 1992; Pilot & Miyazaki 1994; Petersen & Ogawa 2005; Petersen *et al.* 2010) and are accessible electronically through servers maintained at the WHO Collaborating Centers at the Niigata University, Japan and the University of Malmö, Sweden.

Periodontitis "case definition" in epidemiologic studies

A fundamental prerequisite for any meaningful comparative assessment of prevalence is a valid and accurate definition of the disease under investigation.

Unfortunately, no uniform criteria have been established in periodontal research for this purpose. Epidemiologic studies have employed, in an inconsistent manner, a wide array of symptoms, including gingivitis, probing pocket depth, clinical (or probing) attachment level, and radiographically assessed alveolar bone loss. Considerable variation characterizes the threshold values employed for defining periodontal pockets as “deep” or “pathologic”, or the clinical attachment level and alveolar bone scores required for assuming that loss of periodontal tissue support has in fact occurred. In addition, the number of “affected” tooth surfaces required for assigning an individual subject as a “case”, that is as suffering from periodontal disease, has varied. These inconsistencies in the definitions inevitably affect the data describing the distribution of the disease (Papapanou 1996; Kingman & Albandar 2002; Demmer & Papapanou 2010) and, consequently, the identification of risk factors (Borrell & Papapanou 2005). Any review of the literature charged with the task of comparing disease prevalence or incidence in different populations or at different time periods must first confront the interpretation of the published data and literally “decode” them in order to extract relevant information that is amenable to inter-study comparisons. These problems have been addressed in the literature and three specific aspects have attracted special attention, namely (1) the ability of partial recordings to reflect full-mouth conditions, (2) the use of the CPITN system in studies of periodontal disease, and (3) the definition of a “periodontitis case” in epidemiologic studies.

It is clear that an optimal examination of periodontal conditions should include circumferential probing assessments around all teeth. Nevertheless, the majority of epidemiologic studies have, for practical reasons, employed partial recording methodologies. The rationale for the use of partial examinations has been the assumption that (1) the time required for carrying out a partial recording is significantly decreased, resulting in lower cost and better patient acceptance, and (2) the amount of information lost is kept to a minimum, assuming that the examined segments adequately reflect the periodontal condition of the entire dentition. However, attempts to accurately quantify the amount of information lost through the different partial recording systems made by several investigators (Diamanti-Kipioti *et al.* 1993, Eaton *et al.* 2001; Susin *et al.* 2005a; Kingman *et al.* 2008) have revealed that the discrepancy between the findings obtained by means of partial- and full-mouth surveys may be substantial. These studies have typically employed full-mouth data for a series of periodontal parameters and compared them with the values obtained by assessments of a subset of teeth or tooth surfaces. Their results suggest that:

1. High correlations between full-mouth and half-mouth clinical attachment loss scores should be expected in adult populations, due to the apparent

symmetry of periodontal conditions around the midline.

2. The performance of a partial recording system is directly dependent on the actual prevalence and extent of periodontal disease in the population in question and, consequently, on the age of the subjects examined; the less frequent the disease is in the population and the lower the number of sites that are affected in each individual, the more difficult it becomes for the partial examination to accurately portray the full-mouth periodontal status.
3. A full-mouth examination provides the best means of accurately assessing the prevalence and severity of periodontal disease in a population.

The use of the CPITN system in epidemiologic studies of periodontal disease was critically evaluated in a number of publications (Schürch *et al.* 1990; Butterworth & Sheiham 1991; Baelum *et al.* 1993a, b, 1995; Baelum & Papapanou 1996; Benigeri *et al.* 2000). At the time the system was designed, the conversion from periodontal health to periodontitis was thought to follow a continuum of conditions of increasing severity, ranging from health to gingivitis, calculus deposition, formation of deep pockets, and destructive, progressive disease. Consequently, treatment approaches were primarily focused on probing depths to determine the choice between non-surgical and more complex, surgical periodontal therapy. As mentioned earlier, the CPITN system was originally intended for population screening in order to determine treatment needs and to facilitate preventive and therapeutic strategies; it was not meant to describe the prevalence, extent, and severity of periodontal disease and several studies have questioned the suitability of the CPITN for such purposes. For example, Butterworth and Sheiham (1991) examined the ability of CPITN to reflect changes in periodontal conditions in patients of a general dental practice before and after periodontal therapy. Despite a substantial improvement in periodontal status, that is a reduction in gingivitis, calculus scores, and deep pockets, the CPITN scores were only marginally improved. Furthermore, in a rural Kenyan subject sample, Baelum *et al.* (1993b) refuted the validity of the *hierarchical principle* of the CPITN, that is the assumption that a tooth with calculus is assumed to be also positive for bleeding on probing, or that a tooth with deep pockets is assumed to be positive for both calculus and bleeding. In a companion paper, results from a full-mouth examination were compared with those generated by the use of the ten index teeth recommended by the WHO for surveys of adults (Baelum *et al.* 1993a). The study revealed that the partial CPITN methodology seriously underestimated the more severe periodontal conditions both in terms of prevalence and severity, by failing to detect a substantial proportion of subjects with periodontal pockets. Finally, an examination of the relationship

between CPITN findings and the prevalence and severity of clinical attachment loss demonstrated that the CPITN scores do not consistently correlate with clinical attachment loss measures, but tend to overestimate prevalence and severity among younger subjects and underestimate such parameters in elderly populations (Baelum *et al.* 1995). Collectively, the above data call for caution in the interpretation of epidemiologic studies based on the CPITN/CPI systems.

As mentioned earlier, a concise “case definition” is essential for assessing disease prevalence and incidence and to generate comparable data across populations. The currently used classification of periodontal diseases was introduced by the 1999 International Workshop for a Classification of Periodontal Diseases and Conditions and encompasses eight main categories, namely:

- I Gingival diseases
- II Chronic periodontitis
- III Aggressive periodontitis
- IV Periodontitis as a manifestation of systemic diseases
- V Necrotizing periodontal diseases
- VI Abscesses of the periodontium
- VII Periodontitis associated with endodontic lesions
- VIII Developmental or acquired deformities and conditions.

According to the above scheme, the principal forms of periodontitis, which this chapter will mainly focus on, fall under categories II and III, that is *chronic* and *aggressive periodontitis*. *Chronic periodontitis* is the more “common” form that occurs primarily in adults, progresses at a relatively slow rate, resulting in an extent and severity of periodontal tissue loss that is largely commensurate with the presence of local etiologic factors. In contrast, *aggressive periodontitis* is less frequent, affects primarily, but not exclusively, young, systemically healthy individuals, progresses rapidly, and results in substantial loss of periodontal tissue support that may be disproportionate to the occurrence of local etiology. Importantly, *aggressive periodontitis* shows *familial aggregation*, that is it appears to affect parents and siblings within the same family, indicating that genetic predispositions and common environmental exposures are important determinants of the disease. However, none of the three primary features of *aggressive periodontitis* (systemically healthy patient; rapid attachment loss and bone loss; familial aggregation) (Lang 1999) can facilitate the differential diagnosis between *chronic* and *aggressive periodontitis* in the setting of an epidemiologic study: the first because it is entirely non-specific; the second because it requires at least two examinations over time to determine how “rapidly” the periodontal destruction has occurred; and the third because it is subject to *reporting bias*, and requires extensive

interviewing and verification to ascertain reliably. As a result, very sparse epidemiologic data have been generated to date by strictly adhering to the primary criteria of these principal forms of periodontitis. Instead, several studies have reported periodontitis prevalence data using the periodontitis case definition introduced by a working group from the Centers for Disease Control (CDC) and the American Academy of Periodontology (AAP) that is based on a combination of probing depth and clinical attachment level assessments (Page & Eke 2007). The CDC/AAP case definition does not distinguish chronic and aggressive forms of periodontitis, but defines (1) *severe periodontitis* as the presence of two or more interproximal sites with ≥ 6 mm of clinical attachment loss, not on the same tooth, *and* the presence of at least one interproximal site with a ≥ 5 -mm probing depth; and (2) *moderate periodontitis* as the presence of two or more interproximal sites with ≥ 4 mm of clinical attachment loss occurring at two or more different teeth *or* two or more interproximal sites with a ≥ 5 -mm probing depth, not on the same tooth. An alternative, two-level periodontitis case definition for use in epidemiologic studies was developed by a working group of the 5th European Workshop in Periodontology (Tonetti & Claffey 2005), and consisted of a *sensitive* definition (proximal attachment loss of ≥ 3 mm in two or more non-adjacent teeth) and a *specific* definition (proximal attachment loss of ≥ 5 mm in $\geq 30\%$ of the teeth present). The former definition aimed at capturing incipient forms of the disease, while the latter was meant to reflect periodontitis of substantial extent and severity. Lastly, Demmer and Papapanou (2010) recently proposed a method for differentiating chronic and aggressive forms of periodontitis in epidemiologic studies of young individuals (≤ 35 years old) based on a combination of *age* and *extent/severity* of periodontitis. Specifically, these authors suggested that for those aged ≤ 25 years, the presence of two or more interproximal, non-adjacent sites with ≥ 4 mm of attachment loss at a minimum of two different teeth, and accompanied by bleeding on probing, signifies the presence of *aggressive periodontitis*, while a more demanding threshold of attachment loss (≥ 6 mm) is required to arrive at the same diagnosis in individuals aged between 26 and 35 years. In other words, according to this classification approach, extent and severity of periodontal disease exceeding the above thresholds in a young individual is incompatible with a diagnosis of *chronic periodontitis*, as the *de facto* substantial loss of periodontal tissue support early in life indicates rapid progression of the disease and thus justifies its diagnosis as *aggressive periodontitis*. This age-adjusted approach obviously fails to facilitate the differential diagnosis between severe chronic and aggressive periodontitis in a subject over the age of 35 years. Access to disease progression data derived from sequential examinations and/or confirmation of familial aggregation according to the 1999 consensus report are therefore necessary in

order to distinguish the two forms of periodontitis in older patients, but the feasibility of such an approach in the setting of an epidemiologic study remains highly questionable.

Given the lack of universal consensus on the definitions of periodontitis and the continuously evolving epidemiologic approaches, in the following text we have opted to summarize the available data on the prevalence and progression of periodontal disease according to the age range of the examined cohorts. We thus first present findings from epidemiologic studies in adults, including studies exclusively targeting elderly populations, followed by corresponding findings derived from children, adolescents, and young adults. This strategy was adopted under the assumption that the former studies primarily reflect features of *chronic periodontitis*, while the latter studies largely refer to *aggressive periodontitis*.

Prevalence of periodontal diseases

Periodontitis in adults

It is important to briefly mention some older epidemiologic studies to acquire some historical perspective and appreciate how the concepts of both the descriptive and the analytical epidemiology of periodontitis have evolved over the years. Starting with a study performed during the 1950s in India, Marshall-Day *et al.* (1955) used assessments of alveolar bone height to distinguish gingivitis and destructive periodontal disease in 1187 dentate subjects. The authors reported (1) a decrease in the percentage of subjects with “gingival disease without any bone involvement” with increasing age concomitant with an increase in the percentage of subjects with “chronic, destructive periodontal disease”, and (2) a 100% occurrence of destructive periodontitis after the age of 40 years. Findings from other epidemiologic studies from the same period verified a high prevalence of destructive periodontal disease in the adult population in general, and a clear increase in disease prevalence with age. In the 1960s, Scherp (1964) reviewed the available literature on the epidemiology of periodontal disease and concluded that (1) periodontal disease appears to be a major, global public health problem affecting the majority of the adult population after the age of 35–40 years; (2) the disease starts as gingivitis in youth, which, if left untreated, leads to progressive destructive periodontitis; and (3) >90% of the variance of the periodontal disease severity in the population can be explained by age and oral hygiene. These notions, based on established concepts of the pathogenesis of periodontal disease of that time, dominated the periodontal literature until the late 1970s.

Studies performed during the 1980s provided a more thorough description of the site-specific features of periodontal disease and the high variation in periodontal conditions between and within different

populations. Contrary to previous custom, the prevalence issue was no longer addressed through assigning individuals simply to a “periodontitis-affected” or a “disease-free” group, based on presence or absence of attachment or alveolar bone loss. Instead, studies began to unravel details concerning the *extent* to which the dentition was affected by destructive disease (i.e. the percentage of tooth sites involved) and the *severity* of the defects (expressed as the magnitude of the tissue support lost due to the disease). The traditional description of pocket depth and attachment loss scores in terms of *subject mean values* was soon complemented by *frequency distributions*, revealing percentages of tooth sites exhibiting probing depth or attachment level of varying severity. Such an additional analysis appeared necessary after it became clear that mean values offer a crude description of periodontal conditions and fail to reflect the variability in the severity of periodontal disease within and between individuals. In an article presenting different methods of evaluating periodontal disease data in epidemiologic research, Okamoto *et al.* (1988) proposed the use of *percentile plots* in the graphic illustration of attachment loss data. As exemplified by Fig. 7-1, such plots make it possible to illustrate simultaneously both the proportion of subjects exhibiting attachment loss of different levels and the severity of the loss within the subjects. Similar plots

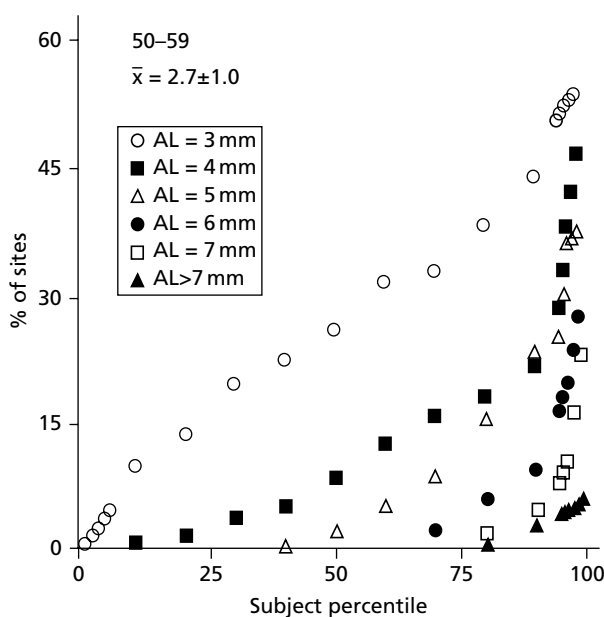


Fig. 7-1 Attachment loss in a group of Japanese subjects aged 50–59 years. The mean value of attachment level and the standard deviation are shown in the top of the figure. The x-axis represents the subject percentile and the y-axis represents the percentage of sites in the subjects showing attachment loss of 3, 4, 5, 6, 7, and >7 mm (represented by 8). Subjects with no or only minor signs of attachment loss are reported to the left and subjects with increasing amounts of periodontal destruction are reported to the right of the graph. For example, the median subject (50th percentile) exhibited 5-mm attachment loss at 2%, 4-mm loss at 8%, and 3-mm loss at 25% of its sites. (Source: Okamoto *et al.* 1988. Reproduced with permission from John Wiley & Sons.)

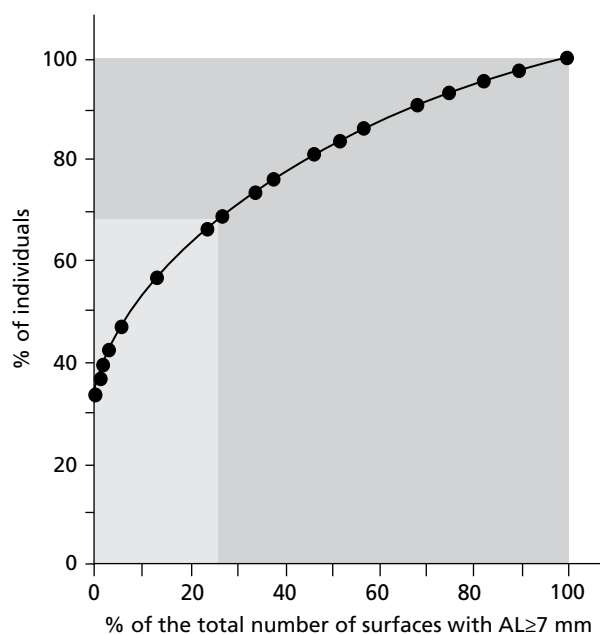


Fig. 7-2 Cumulative distribution of individuals aged ≥ 50 years according to the cumulated proportion of surfaces with attachment loss (AL) of ≥ 7 mm. All individuals are arranged according to increasing number of surfaces with AL of ≥ 7 mm present in each individual. Thus, individuals with few such surfaces are represented on the left side of the diagram and those with many such surfaces on the right side. It is seen that 31% (69–100%) of the individuals account for 75% (25–100%) of the total number of surfaces with AL of ≥ 7 mm present (shaded area). (Source: Baelum *et al.* 1986. Reproduced with permission from John Wiley & Sons.)

may be produced for other parameters, such as gingivitis, probing depths, and gingival recession, and may provide a comprehensive description of both the prevalence and the severity of periodontal disease in a given sample.

Baelum *et al.* (1986) described cross-sectional findings for dental plaque, calculus, gingivitis, loss of attachment, periodontal pockets, and tooth loss in a sample of adult Tanzanians aged 30–69 years. Despite the fact that the subjects examined exhibited large amounts of plaque and calculus, pockets deeper than 3 mm and attachment loss of >6 mm occurred at $<10\%$ of the tooth surfaces. Edentulism was virtually nonexistent, and a very small percentage of subjects had experienced major tooth loss. Of particular interest was the analysis of the distribution of sites within subjects (Fig. 7-2). This analysis revealed that 75% of the tooth sites with attachment loss of >6 mm were found in 31% of the subjects, indicating that a subset of the sample was responsible for the majority of the observed periodontal breakdown. In other words, advanced periodontal disease was not evenly distributed in the population and not readily correlated to supragingival plaque levels; instead, the majority of the subjects examined exhibited negligible periodontal problems, while a limited group was affected by advanced disease.

In a study of similar design performed in Kenya, the same investigators analyzed data from 1131

subjects aged 15–65 years and confirmed their earlier observations (Baelum *et al.* 1988a). Poor oral hygiene in the sample was reflected by high plaque, calculus, and gingivitis scores. However, pockets ≥ 4 mm deep were found in $<20\%$ of the surfaces and the proportion of sites per individual with deep pockets and advanced loss of attachment revealed a pronounced skewed distribution. The authors suggested that “destructive periodontal disease should not be perceived as an inevitable consequence of gingivitis which ultimately leads to considerable tooth loss” and called for a more specific characterization of the features of periodontal breakdown in those individuals who seem particularly susceptible.

At approximately the same time, Löe *et al.* (1986) published data from a longitudinal study that showed distinct patterns for the *progression* of untreated periodontitis. In a population never exposed to any preventive or therapeutic intervention related to oral diseases in Sri Lanka, a cohort of 480 14–31-year-old male tea-plantation laborers was recruited in 1970 and underwent subsequent follow-up examinations. A total of 161 individuals among those originally enrolled were re-examined in 1985, essentially generating data on the natural history of periodontal disease between the ages of 14 and 46 years. Despite poor plaque control and virtually ubiquitous gingival inflammation in the entire sample, three distinct patterns of progression of periodontitis were observed over the follow-up period, based on interproximal longitudinal attachment loss and tooth mortality rates: one group, comprising approximately 8% of the total, exhibited rapid progression of periodontal disease (RP); another group (approximately 11%) exhibited no progression (NP) of periodontal disease beyond gingivitis; and a third group between these two extremes (approximately 81%) exhibited moderate progression (MP). The mean loss of attachment in the RP group was 9 mm and 13 mm at the ages of 35 and 45 years, respectively, as opposed to 1 mm and 1.5 mm in the NP group, and 4 mm and 7 mm in the MP group. As a result, the annual rate of longitudinal attachment loss in the RP group varied between 0.1 and 1.0 mm, in the MP group between 0.05 and 0.5 mm, and in the NP group between 0.05 and 0.09 mm. Thus, this study clearly demonstrated huge variability in progression of periodontitis in a seemingly homogeneous population, and suggested that variables other than age, plaque, and gingival inflammatory status are important determinants of periodontal deterioration over time.

Several more recent epidemiologic studies verify the above principals. In these studies, periodontal disease has been assessed by means of clinical examination of the periodontal tissues (Brown *et al.* 1990; Albandar *et al.* 1999; Susin *et al.* 2004a; Thomson *et al.* 2006); radiographic assessments of alveolar bone loss (Papapanou *et al.* 1988; Jenkins & Kinane 1989; Salonen *et al.* 1991; Diamanti-Kipioti *et al.* 1995); or a combination of clinical and radiographic means (Papapanou *et al.* 1990; Hugoson *et al.* 1992, 1998a, 2005).

Table 7-1 Selected prevalence studies of periodontitis in adults.

Authors/country	Sample/methodology	Findings
Løe <i>et al.</i> (1978) Norway/Sri Lanka	Two samples, one comprising 565 Norwegian students and academicians and the other 480 Sri Lankan tea laborers, in ages 16–30+ years; assessments of plaque, gingivitis, calculus, PD, and AL at the mesial and facial aspects of all teeth	Norwegian group: excellent oral hygiene, negligible amounts of plaque and gingivitis, virtually no deep pockets and minimal attachment loss; mean AL at the age of 30 years <1 mm Sri Lankan group: poor oral hygiene, abundant plaque and calculus, attachment loss present at the age of 16 years, increasing with age; mean AL at the age of 30 years ≈ 3 mm, a substantial number of teeth with AL of >10 mm
Baelum <i>et al.</i> (1988a) Kenya	A stratified random sample of 1131 subjects, 15–65 years; full-mouth assessments of tooth mobility, plaque, calculus, bleeding on probing (BoP), PD, and AL	Plaque in 75–95% and calculus in 10–85% of all surfaces PD ≥4 mm in <20% of sites AL of ≥1 mm in 10–85% of sites Percentage of sites/subjects with PD or AL of ≥4 mm or ≥7 mm conspicuously skewed
Brown <i>et al.</i> (1990) USA	A sample of 15 132 subjects, stratified by geographic region, representing 100 million employed adults aged 18–64 years; probing assessments at mesial and buccal sites in one upper and one lower quadrant; mesial assessments performed from the buccal aspect of the teeth; assessments of gingivitis, PD, AL, and gingival recession	44% of all subjects had gingivitis at an average of 2.7 sites/subject and at <6% of all sites assessed Pockets 4–6 mm were observed in 13.4% of subjects at an average of 0.6 sites/person and at 1.3% of all sites assessed; corresponding figures for pockets ≥7 mm were 0.6%, 0.01, and 0.03% AL ≥3 mm was prevalent in 44% of subjects (increasing with age from 16% to 80%) affecting an average of 3.4 sites/subject; corresponding figures for AL ≥5 mm were 13% (2–35%) and 0.7 sites/subject
Salonen <i>et al.</i> (1991) Sweden	A random sample of 732 subjects, 20–80+ years, representing 0.8% of the population of a southern geographic region; full-mouth radiographic examination; alveolar bone level expressed as a percentage of the root length (B:R ratio); B:R of ≥80% represents intact periodontal bone support	Age group of 20–29 years: 38% of subjects had no sites with a B:R of <80% and 8% of subjects had ≥5 sites below this threshold Corresponding figures for the age group 50–59 years: 5% and 75%; after the age of 40, women displayed more favorable B:R ratios than men
Hugoson <i>et al.</i> (1998) Sweden	Three random samples of 600, 597, and 584 subjects aged 20–70 years, examined in 1973, 1983, and 1993, respectively; full-mouth clinical and radiographic examination; based on clinical and radiographic findings, the subjects were classified according to severity of periodontal disease in five groups, where group 1 (G1) included subjects with close to faultless periodontal tissues and group 5 (G5) subjects with severe disease	Edentulism decreased over the 20-year period from 11% to 8% to 5%; percentage distribution of the subjects in the five groups in 1973, 1983, and 1993 respectively was: G1 8%, 23%, 22%; G2 41%, 22%, 38%; G3 47%, 41%, 27%; G4 2%, 11%, 10%; G5 1%, 2%, 3%; the increase in the prevalence of subjects with severe disease was apparently due to the increased number of dentate subjects at the older ages
Albandar <i>et al.</i> (1999) USA	A nationally representative, multistage probability sample comprising 9689 subjects, 30–90 years old (NHANES III study); probing assessments at mesial and buccal sites in one upper and one lower quadrant; mesial assessments performed on the buccal aspect of the teeth; assessments of gingivitis, PD, and location of the gingival margin in relation to the CEJ	Pockets ≥5 mm were found in 8.9% of all subjects (7.6% in non-Hispanic Caucasians, 18.4% in non-Hispanic Blacks, and 14.4% in Mexican Americans) AL ≥5 mm occurred in 19.9% of all subjects (19.9% in non-Hispanic Caucasians, 27.9% in non-Hispanic Blacks, and 28.34% in Mexican Americans)
Schürch & Lang (2004) Switzerland	A total of 1318 subjects, randomly selected based on community rosters in seven regions, aged 20–89 years; probing assessments of PD and AL for all teeth present; assessments of plaque and gingivitis for index teeth	7.1% of subjects were edentulous; mean number of teeth present in dentate subjects was 21.6 Mean values of PD reached a plateau of 3 mm by the age of 49 years AL increased dramatically after the age of 50 years and paralleled a marked loss of teeth
Susin <i>et al.</i> (2004a) Brazil	A sample of 853 dentate individuals, selected by multistage probability sampling, aged 30–103 years; full-mouth examination of AL at six sites/tooth	Moderate AL (≥5 mm) and advanced AL (≥7 mm) occurred in 70% and 52% of the subjects, affecting an average of 36% and 16% of their teeth, respectively; in comparison to 30–39 year olds, 40–49 year olds had 3x increased risk for moderate and 7.4x increased risk for advanced AL; corresponding figures for ≥50 year olds were 5.9x and 25.4x, respectively

Table 7-1 Continued.

Authors/country	Sample/methodology	Findings
Dye <i>et al.</i> (2007) USA	NHANES 1999–2004 study, nationally representative sample comprising 10 312 individuals in four age cohorts (35–49, 50–64, 65–74, and 75+ years; partial-mouth examination in two randomly selected quadrants (one maxillary and one mandibular) at the mesio-facial and mid-facial sites of all fully erupted teeth excluding third molars	Prevalence of AL ≥ 3 mm in the four age cohorts was 36.1%, 53.4%, 67.2%, and 75.5% respectively Corresponding figures for PD ≥ 4 mm were 11.9%, 13.2%, 11.3%, and 12.1% respectively
Wang <i>et al.</i> (2007) China	A sample of 1590 dentate subjects with ≥ 14 teeth present, aged >25 years, from four geographic regions, equally farmers and urban professionals; partial-mouth examination at six sites in each of six index teeth	Average of 40% of sites bled on probing in the rural group as compared to 35% in the urban group Prevalence of AL ≥ 4 mm was approximately 10% in ages 25–34 years, increasing to 31%, 53%, and 70% in ages 35–44, 45–59, and >60 years, respectively, in the rural group; corresponding figures were 18%, 38%, and 57% in the urban group
Holtfreter <i>et al.</i> (2010) Germany	The fourth German Dental Health Survey examined a total of 1965 individuals aged 35–44 years (adult sample) and 65–74 years (senior sample); partial-mouth examination of PD and AL at three sites in each of 12 index teeth	AL ≥ 3 mm was prevalent in 95% of the adults and 99.2% of the seniors (68.7% and 91.4% of teeth affected, respectively) PD ≥ 4 mm was prevalent in 70.9% and 87.4% of the adult and senior cohorts, respectively
Eke <i>et al.</i> (2012) USA	NHANES 2009–2010 study, nationally representative sample comprising 3742 individuals in four age cohorts (30–34, 35–49, 50–64, 65+ years); full-mouth examination at six sites per tooth at all fully erupted teeth excluding third molars	Prevalence of AL ≥ 3 mm in the four age cohorts was 64.1%, 83.1%, 92.0%, and 96.7%, respectively Corresponding figures for PD ≥ 4 mm were 29.6%, 35.5%, 47.5%, and 49.3%, respectively Prevalence of severe periodontitis, according to the CDC/AAP definition (Page & Eke 2007) was 1.9% in ages 30–34 years, 6.7% in ages 35–49 years, 11.7% in ages 50–64 years, and 11.2% in ages 65 years and above

PD, probing depth; AL, attachment level; CEJ, cemento-enamel junction; NHANES, National Health and Nutrition Examinations Surveys; CDC/AAP, Centers for Disease Control/American Academy of Periodontology.

Table 7-1 summarizes the design and main findings from a number of cross-sectional studies in adults from geographically divergent areas and involving samples of a relatively large size. Several included studies focused on assessments of prevalence of “advanced periodontitis”, the definition of which is, however, far from identical across the different reports, rendering comparisons difficult. Nevertheless, it appears that severe forms of periodontitis affect a relatively limited portion of the population in the industrialized countries, usually not exceeding 10–15%. This fraction increases considerably with age and appears to reach its peak at the age of 50–60 years. The increased loss of periodontally affected teeth occurring after this age appears to account for the subsequent decline in prevalence. It is worth pointing out that studies employing full-mouth examination protocols generally generate higher prevalence estimates, underscoring the decisive impact of the methodology. An interesting issue is also the apparent disparity in the extent and severity of periodontitis across different populations. In an earlier report by Baelum *et al.* (1996), the authors recalculated their own data from a Kenyan (Baelum *et al.* 1988a) and a Chinese (Baelum *et al.* 1988b) adult population to conform with the examination

methodology and data presentation utilized in each of six other surveys: from Japan (Yoneyama *et al.* 1988), Norway (Löe *et al.* 1978), New Mexico (Ismail *et al.* 1987), Sri Lanka (Löe *et al.* 1978), and two South Pacific islands (Cutress *et al.* 1982). Among the samples included in this analysis, only the Sri Lankan and the South Pacific island subjects appeared to suffer severe periodontal tissue breakdown, while the distribution of advanced disease was strikingly similar in six of the eight samples, despite marked differences in oral hygiene conditions. Hence, the data failed to corroborate the traditional generalization that the prevalence and severity of periodontitis is markedly increased in African and Asian populations. On the other hand, data from the National Health and Nutrition Examinations Surveys (NHANES) (Albandar *et al.* 1999; Dye *et al.* 2007; Eke *et al.* 2012), which examined large nationally representative, stratified, multistage probability samples in the US, clearly showed that the prevalence of deep pockets and advanced attachment loss was more pronounced in non-Hispanic black than in non-Hispanic white subjects. This observation was consistent even when several alternative thresholds defining advanced disease were employed. Thus, current evidence suggests that the prevalence of severe periodontitis is not

Table 7-2 Selected prevalence studies of periodontitis in elderly subjects.

Authors/country	Sample/methodology	Findings
Baelum <i>et al.</i> (1988b) China	544 persons, aged 60+ years, from two urban and one rural area of the Beijing area; assessments of plaque, calculus, gingivitis, loss of attachment, pocket depth and tooth mobility	0–29% edentulous; mean number of teeth 6.9–23.9, depending on age and sex ≈50% of all surfaces with plaque and calculus 50% of all sites with AL of ≥4 mm <15% with PD ≥4 mm Conspicuously skewed percentage of sites/persons with AL of ≥7 mm and PD ≥4 mm
Locker & Leake, (1993a) Canada	907 subjects, aged 50–75+ years, living independently in four communities; probing assessments at mesio-buccal and mid-buccal aspects of all teeth; mid-palatal and mesio-palatal probing assessments in upper molars; 23% of subjects edentulous; calculation of Extent and Severity Index (ESI) with AL threshold set at ≥2 mm; “severe disease”: >4 sites with AL ≥5 mm and PD ≥4 mm at ≥1 of those sites	59% of subjects with PD of ≥4 mm, 16% with ≥6 mm, and 3% with ≥8 mm 86% of subjects with AL of ≥4 mm, 42% with ≥6 mm, and 16% with ≥8 mm; 20% of the subjects with a mean PAL of ≥4 mm Severe disease in 22% of subjects; mean ESI: 77, 2.44
Beck <i>et al.</i> (1990) USA	690 community dwelling adults, aged 65+ years; probing assessments at mesio-buccal and mid-buccal surfaces, all teeth; “advanced disease”: ≥4 sites with AL of ≥5 mm and PD ≥4 mm at ≥1 of those sites	Mean ESI in Blacks: 78, 4; in Caucasians: 65, 3.1 Advanced disease in 46% of Blacks and 16% of Caucasians
Gilbert & Heft (1992) USA	671 dentate subjects, aged 65–97 years, attending senior activity centers; probing assessments at mesial and buccal surfaces of one upper and one lower quadrant; questionnaire data; calculation of ESI	Average of 17.0 teeth/subject 50.7% of subjects with most severe mesial PD of 4–6 mm and 3.4% with PD of ≥7 mm 61.6% with most severe AL of 4.6 mm and 24.2% with AL of ≥7 mm ESI increased with age: 84.8, 3.6 (65–69 years); 88.7, 3.8 (75–79 years); 91.2, 3.9 (85+ years)
Douglass <i>et al.</i> (1993) USA	1151 community-dwelling elders, aged 70+ years; probing assessments at ≥3 sites/tooth, all teeth; 57% of the sample female, predominantly Caucasian (95%); 37.6% edentulous; mean number of teeth present between 21.5 and 17.9, depending on age	85% of subjects with BOP 66% with 4–6-mm deep pockets affecting an average of 5.3 teeth/subject; 21% with pockets of >6 mm affecting an average of 2.2 teeth 39% with AL of 4–6 mm at 6.7 sites/subject and 56% with AL of >6 mm at 2.7 teeth/subject
Kiyak <i>et al.</i> (1993) USA	1063 residents in 31 nursing homes, aged 72–98 years; visual inspection of the oral cavity; periodontal status assessed indirectly through registration of intraoral swelling or suppuration, sore or bleeding gums, increased tooth mobility, and poor oral hygiene	42% of subjects with remaining natural teeth; 43% had sore or bleeding gums, 18% with significant tooth mobility, 6% with intraoral swelling or suppuration, and 72% with poor oral hygiene
Weyant <i>et al.</i> (1993) USA	650 long-term residents of nursing home care units, mean age 72 years; probing assessments at mesial and buccal surfaces, all teeth; demographic, oral and general health data recorded; sample predominantly male and Caucasian; calculation of ESI scores	42% edentulous 60% of subjects with PD of >3 mm at an average of 5.8 sites/person; 3.7% with PD of ≥6 mm at <1 site/person Overall mean mesial ESI of 74, 2.91
Bourgeois <i>et al.</i> (1999) France	603 non-institutionalized elderly, aged 65–74 years; stratified sample with respect to gender, place of residence and socioeconomic group; periodontal conditions assessed by CPITN	16.3% edentulous 31.5% of subjects had pockets ≥4 mm; 2.3% had pockets ≥6 mm
Levy <i>et al.</i> (2003) USA	From a sample of 449 community dwelling elders, mean age 85 years, 342 (76%) were dentate and 236 were examined with respect to PD and AL at four sites/tooth in all teeth present	91% of subjects had ≥1 site with ≥4 mm AL, 45% ≥1 site with ≥6 mm AL, and 15% ≥1 site with ≥8 mm AL
Mack <i>et al.</i> (2004) Germany	1446 randomly selected subjects aged 60–79 years; half-mouth examination of PD and AL at four sites/ tooth; plaque calculus and BoP were assessed at index teeth	16% of the 60–65-year olds and 30% of the 75–79-year olds were edentulous Among the 70–79-year olds, median BoP was 37.5% in men and 50% in women Prevalence of PD ≥6 mm was 31.8% and 28.5% in men and women, respectively Prevalence of AL ≥5 mm was 71.9% and 66.9% in men and women, respectively
Syrjälä <i>et al.</i> (2010) Finland	1460 individuals aged ≥65 years, participants in the nationally representative Health 2000 Survey; full-mouth examination at four sites/tooth at all erupted teeth except third molars	44.3% edentulous 31% of dentate participants had no pockets >3 mm; 28% had 1–3 teeth with ≥4 mm pockets, 15% had 4–6 and 26% ≥7 affected teeth 73% showed BoP at ≥1 sextant

PD, probing depth; AL, attachment level; BoP, bleeding on probing; CEJ, cemento-enamel junction; ESI, Extent and Severity Index; CPITN, Community Periodontal Index of Treatment Needs.

uniformly distributed among various races, ethnicities or socioeconomic groups.

The limitations of the findings from studies using the CPITN system were discussed earlier. However, a substantial part of the available information from developing countries has been collected using this particular system. An article providing a summary of almost 100 CPITN surveys from >50 countries performed over the period 1981–89 for the age group 35–44 years was published by Miyazaki *et al.* (1991b). These studies indicate a huge variation in the percentage of subjects with one or several deep (≥ 6 mm) pockets, both between and within different geographic areas. The percentage of subjects with such deep pockets ranged between 1% and 74% in Africa (data from 17 surveys), 8% and 22% in North and South America (four surveys), 2% and 36% in the Eastern Mediterranean (six surveys), 2% and 40% in Europe (38 surveys), 2% and 64% in South-East Asia, and 1% and 22% in the Western Pacific area (17 surveys). The average number of sextants per subject with ≥ 6 -mm deep pockets also varied considerably and ranged between 0 and 2.1 in Africa, 0.1 and 0.4 in North and South America, 0.1 and 0.6 in the Eastern Mediterranean, 0.1 and 0.8 in Europe, 0.1 and 2.1 in South-East Asia, and 0 and 0.4 in the Western Pacific area. However, it is difficult to assess the extent to which these values reflect true differences in the periodontal conditions given the methodologic limitations of the CPITN system.

Table 7-2 summarizes a number of prevalence studies of periodontal disease in elderly subjects. In four studies (Beck *et al.* 1990; Gilbert & Heft 1992; Locker & Leake 1993b; Weyant *et al.* 1993), data on attachment loss were used to calculate ESI scores, which appear to be relatively consistent between the surveys. It is evident that attachment loss of moderate magnitude was frequent and widespread in these subject samples; however, severe disease was again found to affect relatively limited proportions of the samples and usually only a few teeth per subject. It must be realized, however, that (1) edentulism is high in elderly subjects, and (2) the remaining teeth in elderly individuals are likely those least affected by periodontitis. As discussed later, tooth loss results in an underestimation of the “true” extent and severity of periodontitis in elderly individuals.

Periodontitis in children and adolescents

The form of periodontal disease that affects the primary dentition, the condition formerly termed *prepubertal periodontitis*, has been reported to appear in both a generalized and a localized form (Page *et al.* 1983). Information about this disease has mainly been provided by clinical case reports and no data on the prevalence and the distribution of the disease in the general population are available. However, a small number of studies involving samples of children

have provided limited data on the frequency with which deciduous teeth may be affected by attachment loss. The criteria used in these studies are by no means uniform; hence the prevalence data vary significantly. In an early study, Jamison (1963) examined the “prevalence of destructive periodontal disease” (indicated by PDI scores > 3) in a sample of 159 children in Michigan, USA and reported prevalences of 27% for 5–7-year-old children, 25% for 8–10-year olds, and 21% for 11–14-year olds. Shlossman *et al.* (1986) used an attachment level value of ≥ 2 mm as a cut-off point and reported prevalences of 7.7% in 5–9-year olds and 6.1% in 10–14-year olds in a sample of Pima Indians. Sweeney *et al.* (1987) examined radiographs obtained from 2264 children, aged 5–11 years, who were referred to a university clinic for routine dental treatment, and reported that a distinct radiographic bone loss was evident at one or more primary molars in 19 children (0.8%), 16 of whom were black, two Caucasian, and one Asian.

In contrast, relatively uniform criteria have been used in epidemiologic studies of periodontitis in teenagers and young adults, although, as discussed earlier, the vast majority of studies have not employed the diagnostic criteria of the current classification. Nevertheless, the clinical phenotype of *localized aggressive periodontitis* (LAP) in young individuals, the condition formerly termed *localized juvenile periodontitis* (LJP), that is the severe and disproportionate involvement of incisors and first molars, has facilitated more standardized detection methodologies. Typically, a two-stage approach has been adopted in these studies: first, bitewing radiographs are used to screen for bone loss adjacent to molars and incisors, and then a clinical examination is performed to verify the diagnosis. As illustrated by the data in Table 7-3, the prevalence of LAP varies in geographically and/or racially different populations. In Caucasians, the disease appears to affect females more frequently than males and the prevalence is low (approximately 0.1%). In other races, and in particular in Blacks, the disease is more prevalent, probably at levels over 1%, and the sex ratio appears to be reversed, since males are affected more frequently than females. Smoking and low socioeconomic status have been confirmed to be associated with *aggressive periodontitis* in various populations (Lopez *et al.* 2001; Susin & Albandar 2005; Levin *et al.* 2006).

Epidemiologic studies of periodontal conditions in adolescents have been also carried out using the CPITN system. Miyazaki *et al.* (1991a) presented an overview of 103 CPITN surveys of subjects aged 15–19 years from over 60 countries. The most frequent finding in these groups was the presence of calculus, which was much more prevalent in subjects from non-industrialized than industrialized countries. Probing pocket depths of 4–5 mm were present in about two-thirds of the populations examined. However, the occurrence of deep pockets (≥ 6 mm) was relatively infrequent: score 4 quadrants were

Table 7-3 Selected prevalence studies of periodontitis in adolescents and young adults.

Authors/Country	Sample/methodology	Findings
Saxén (1980) Finland	A random sample of 8096 16-year olds; radiographic and clinical criteria (bone loss adjacent to first molars without any obvious iatrogenic factors and presence of pathologic pockets)	Prevalence of LAP 0.1% (8 subjects, 5 of whom were females)
Kronauer <i>et al.</i> (1986) Switzerland	A representative sample of 7604 16-year olds; two step examination (radiographic detection of bone lesion on bite-wing radiographs, clinical verification of presence of pathologic pockets)	Prevalence of LAP of 0.1%; 1:1 gender ratio
Saxby (1987) UK	A sample of 7266 schoolchildren; initial screening by probing assessments around incisors and first molars; LAP cases diagnosed definitively by full-mouth clinical and radiographic examination	Overall prevalence of LAP of 0.1%, 1:1 gender ratio; however, prevalence varied in different ethnic groups (0.02% in Caucasians, 0.2% in Asians, and 0.8% in Afro-Caribbeans)
Neely (1992) USA	1038 schoolchildren 10–12 years old, volunteers in a dentifrice trial; three-stage examination including radiographic and clinical assessments; bite-wing radiographs screened for possible cases; bone loss measurements of the CEJ–bone crest distance of ≥ 2 mm used to identify probable cases; LAP diagnosed clinically as PD of ≥ 3 mm at ≥ 1 first permanent molars in the absence of local irritants	117 possible and 103 probable cases identified in steps 1 and 2, respectively; of 99 probable cases contacted, 43 were examined clinically; 2 cases of LAP confirmed in stage 3, yielding a prevalence rate of 0.46%
Cogen <i>et al.</i> (1992) USA	4757 children, aged <15 years, from the pool of a children's hospital; retrospective radiographic examination of two sets of bite-wings; LAP diagnosed in case of arc-shaped alveolar bone loss in molars and/or incisors	Caucasians: LAP prevalence 0.3%, female:male ratio 4:1 Blacks: LAP prevalence 1.5%, female:male ratio $\approx 1:1$ Among Black LAP cases with available radiographs from earlier examinations, 85.7% showed evidence of bone loss in the mixed dentition and 71.4% in the deciduous dentition
Löe & Brown (1991) USA	National Survey of US children, multistage probability sampling representing 45 million schoolchildren; 40 694 subjects, 14–17 years old examined; probing assessments at mesial and buccal sites, all teeth; LAP: ≥ 1 first molar and ≥ 1 incisor or second molar and ≤ 2 cuspids or premolars with ≥ 3 mm AL; GAP: if LAP criteria not met and ≥ 4 teeth (of which ≥ 2 were second molars, cuspids or premolars) with ≥ 3 mm AL; incidental loss of attachment (ILA): if neither LAP nor GAP criteria met but ≥ 1 teeth with ≥ 3 mm AL; bivariate and multivariate analysis	Population estimates: LAP 0.53%; GAP 0.13%; ILA 1.61%; all 2.27% representing almost 300 000 adolescents; Blacks at much higher risk for all forms of early-onset disease than Caucasians Males more likely (4.3:1) to have GAP than females, after adjusting for other variables; Black males 2.9 times as likely to have LAP than Black females; Caucasian females more likely to have LAP than Caucasian males by the same odds
Bhat (1991) USA	A sample of 11 111 schoolchildren, 14–17 years old; probing assessments at mesial and buccal surfaces of all teeth; multistage cluster sampling stratified by age, sex, seven geographic regions, and rural or urban residence; not stratified by race or ethnicity	22% of children with ≥ 1 site with AL of ≥ 2 mm, 0.72% of ≥ 4 mm, and 0.04% of ≥ 6 mm Supra- and sub-gingival calculus in 34% and 23% of children, respectively
van der Velden <i>et al.</i> (1989) The Netherlands	4565 subjects, 14–17 years old examined; randomization among high school students; probing assessments at the mesio- and disto-facial surfaces of first molars and incisors; one bacterial sample from the dorsum of the tongue and one subgingival plaque sample from the site with maximal attachment loss obtained from 103 of the 230 subjects with AL and cultured for identification of <i>Aggregatibacter actinomycetemcomitans</i>	Overall, AL occurred in 5% of the sample and was more frequent in males; 16 subjects (0.3%) had ≥ 1 site with AL of 5–8 mm; female:male ratio in this group 1.3:1 <i>A. actinomycetemcomitans</i> identified in 17% of the sampled subjects with AL
Lopez <i>et al.</i> (1991) Chile	2500 schoolchildren in Santiago (1318 male, 1182 female), aged 15–19 years; clinical and radiographic assessments; three-stage screening: (1) clinical assessments of PD at incisors and molars, (2) children with ≥ 2 teeth with PD of ≥ 5.5 mm subjected to a limited radiographic examination, and (3) children with alveolar bone loss of ≥ 2 mm invited for a full-mouth clinical and radiographic examination	After screening, 27 subjects had a tentative diagnosis of LAP, of which 8 were confirmed (7 female, 1 male); overall prevalence of LAP 0.32%, 95% CI 0.22–0.42%; LAP significantly more frequent in the low socioeconomic group

Table 7-3 Continued.

Authors/Country	Sample/methodology	Findings
Ben Yehouda <i>et al.</i> (1991) Israel	1160 male Israeli army recruits, aged 18–19 years; panoramic radiography; JP diagnosed on the basis of bone loss involving $\geq 30\%$ of the root length adjacent to first molars or incisors	10 recruits (0.86%, 95% CI 0.84–0.88%) had a bone loss pattern consistent with localized juvenile periodontitis
Melvin <i>et al.</i> (1991) USA	5013 military recruits, aged 17–26 years; panoramic radiography followed by full-mouth clinical examination; diagnosis of JP if bone loss and attachment loss was greater at first molars and/or incisors than at other teeth	Overall prevalence of JP 0.76%, female:male ratio 1.1:1 Prevalence in Blacks: 2.1%, female:male ratio 0.52:1 Prevalence in whites: 0.09%, female:male ratio 4.3:1
Tinoco <i>et al.</i> (1997) Brazil	7843 schoolchildren, aged 12–19 years; two-stage screening: (1) clinical assessment of PD at first molars, (2) children with ≥ 1 tooth with PD ≥ 5 mm examined further; LAP diagnosed if a person with no systemic disease presented with AL > 2 mm at ≥ 1 sites with radiographic evidence of bone loss and ≥ 1 infrabony defects at molars/incisors	119 subjects identified at initial screening; 25 confirmed cases of LAP; overall prevalence 0.3% Ethnic origins and gender ratios not reported
Lopez <i>et al.</i> (2001) Chile	A random sample of 9162 high school students, aged 12–21 years; probing assessments of AL at six sites/tooth at all incisors and molars	Prevalence of AL of ≥ 1 mm was 69.2%, of ≥ 2 mm was 16%, and of ≥ 3 mm was 4.5%. AL was associated with older age, female gender, poor oral hygiene, and lower socioeconomic status
Levin <i>et al.</i> (2006) Israel	642 army recruits (87.5% men), aged 18–30 years (mean 19.6 years); radiographic and clinical examination of first molars and incisors	AP prevalence was 5.9% (4.3% LAP, 1.6% GAP); current smoking and north African origin were significantly related to AP
Taylor & Borgnakke (2007) USA	105 young adults aged 18–25 years, living in housing units in Detroit, Michigan; full-mouth examination at four sites/tooth at all teeth present	9% and 2% of sample were found to suffer from “moderate” or “severe” periodontitis, respectively, according to the CDC/AAP criteria (Page & Eke 2007)
Holtfreter <i>et al.</i> (2009) Germany	587 young adults, aged 20–29 years, participants in the Study of Health in Pomerania (SHIP); half-mouth examination of an upper and lower quadrant at four sites/tooth with respect to PD and AL	12% and 1% of sample were found to suffer from “moderate” or “severe” periodontitis, respectively, according to the CDC/AAP criteria, as above 5% of sample exhibited AL ≥ 4 mm, 2% ≥ 5 mm and 1% ≥ 6 mm

Terms used in all but Levin *et al.* (2006): “localized juvenile periodontitis (JP)” instead of “localized aggressive periodontitis (LAP)”, and “generalized juvenile periodontitis” instead of “generalized aggressive periodontitis (GAP)”.

PD, probing depth; AL, attachment level; CEJ, cemento-enamel junction; AP, aggressive periodontitis; CDC/AAP, Centers for Disease Control/American Academy of Periodontology; CI, confidence interval.

reported to occur in only ten of the examined populations (in four of the nine examined American samples, one of 16 African samples, one of ten Eastern Mediterranean samples, two of 35 European samples, two of 15 South-East Asian samples, and none of 18 Western Pacific samples).

The progression pattern of periodontitis in a sample of 167 adolescents in the UK was studied in a 5-year longitudinal study by Clerehugh *et al.* (1990). In this study, 3% of the initially 14-year olds had attachment loss of ≥ 1 mm affecting $> 1\%$ of sites. However, at age 19 years, 77% showed a similar level of attachment loss and 31% of sites were affected. Presence of subgingival calculus at baseline was significantly associated with disease progression. In a study involving a larger sample size in the US, Brown *et al.* (1996) studied a nationally representative sample comprising 14 013 adolescents with respect to the pattern of progression of the disease entity formerly

termed *early-onset periodontitis*, that is the type of periodontitis that occurs in individuals of a young age. Subjects were diagnosed at baseline as free from periodontitis, or suffering from LAP, *generalized aggressive periodontitis* (GAP), or incidental attachment loss (IAL). Of the individuals diagnosed with LAP at baseline, 62% continued to display localized periodontitis lesions 6 years later, but 35% developed a generalized disease pattern. Among the group initially diagnosed as suffering from IAL, 28% developed LAP or GAP, while 30% were reclassified in the no attachment loss group. Molars and incisors were the teeth most often affected in all three affected groups. Thus, the study indicated that these three forms of periodontitis may progress in a similar fashion, and that certain cases of LAP may develop into a more generalized form of *aggressive periodontitis*.

The possibility that LAP and *prepubertal periodontitis* are associated conditions, that is the former is a

development of the latter, has also attracted attention. In an early study, Sjödin *et al.* (1989) retrospectively examined radiographs of the primary dentition of 17 subjects with LAP and reported that 16 of these subjects showed a CEJ–bone crest distance of ≥ 3 mm in at least one tooth site in their deciduous dentition. The same research group (Sjödin & Matsson 1992) examined the CEJ–bone crest distance on radiographs from 128 periodontally healthy children aged 7–9 years, in order to define a threshold value that, if exceeded, would indicate a high probability of periodontal pathology around the deciduous teeth. Having set this threshold value at 2 mm, Sjödin *et al.* (1993) retrospectively examined radiographs of the deciduous dentition from 118 patients with *aggressive periodontitis* and 168 age- and sex-matched periodontally healthy controls. The patients were divided in two groups, one comprising those with only one affected site (45 subjects) and another (73 subjects) including those with 2–15 sites with bone loss in their permanent dentition. It was found that 52% of the patients in the latter group, 20% of those in the former group, and only 5% of the controls exhibited at least one site with bone loss in their primary dentition. The authors concluded that, at least in some young subjects with *aggressive periodontitis*, the onset of the disease may manifest in the primary dentition. Similar results were reported by Cogen *et al.* (1992) from a study in the US. Among systemically healthy young Black subjects with *aggressive periodontitis* and available radiographs of the primary dentition, 71% showed alveolar bone loss adjacent to one or several primary teeth. Finally, an interesting radiographic study of the mixed dentition in Australian children aged 5–12 years by Darby *et al.* (2005) investigated the prevalence of alveolar bone loss around first permanent molars, and first and second deciduous molars. Based on radiographs of 542 children, 13% were found to display definite bone loss, that is bone levels >3.0 mm from the CEJ. Half of all sites with definite bone loss were on the second deciduous molars and, in the vast majority, on distal tooth surfaces. In other words, this study showed that the tooth surface of the deciduous dentition most frequently affected by bone loss was the one in close proximity to the most frequent localization of *aggressive periodontitis* in older children with permanent dentition, that is the mesial surface of the first permanent molar.

Periodontitis and tooth loss

Tooth loss may be the ultimate consequence of destructive periodontal disease. Teeth lost due to the sequelae of the disease are obviously not amenable to registration in epidemiologic surveys and may, hence, lead to an underestimation of the prevalence and the severity of the disease. The well-established epidemiologic concept of *selection bias* (also referred to as the *healthy survivor effect*, indicating that the comparatively healthier subjects will present for an examination while the more severely affected may

refuse participation or fail to present because of the morbidity itself) is in this context applicable at the individual tooth level, since the severely affected teeth may have already been extracted/lost. Aspects related to tooth loss on a population basis have been addressed in numerous publications. Important questions that were analyzed included the relative contribution of periodontitis to edentulism (Eklund & Burt 1994; Takala *et al.* 1994) or to tooth extractions in subjects retaining a natural dentition (Reich & Hiller 1993; McCaul *et al.* 2001; Susin *et al.* 2005b; Thorstensson & Johansson 2010; Hirotsu *et al.* 2011).

Typically, surveys addressing the first topic have utilized questionnaire data obtained from general practitioners instructed to document the reasons why teeth were extracted over a certain time period. The results indicate that the reason underlying the vast majority of extractions in ages up to 40–45 years is dental caries. However, in older age cohorts, periodontal disease is about equally responsible for tooth loss. Overall, periodontitis is thought to account for 30–35% of all tooth extractions, while caries and its sequelae for up to 50%. In addition, caries appears to be the principal reason for extractions in cases of total tooth clearance. Finally, identified risk factors for tooth loss include smoking, poor dental health, poverty and other socio-behavioral traits, and poor periodontal status.

Obviously, it is not feasible to “translate” tooth loss data into prevalence figures of periodontal disease. An evaluation, however, of the periodontal status at the population level, and in particular in older age cohorts, must weigh the information provided by tooth loss data; otherwise underestimation of the occurrence and the sequelae of the disease is inevitable (Gilbert *et al.* 2005).

Risk factors for periodontitis

Introduction: Definitions

There is an abundance of both empirical evidence and substantial theoretical justification for accepting the widespread belief that many diseases have more than one cause, in other words they have a *multifactorial etiology* (Kleinbaum *et al.* 1982). Consequently, in any particular instance when a *causal relationship* is investigated, the specificity of the relation between exposure to an etiologic agent and effect (the *necessity* or the *sufficiency* of the condition) may be challenged. In the case of most infectious diseases for example, it is known that the presence of the microbial agent (which is defined as the necessary condition) is not always accompanied by signs or symptoms characteristic of that disorder. Thus, the agent itself is not sufficient to cause any pathologic occurrence; rather, the disease development may be dependent on multiple and diverse additional factors, including specific host responses, toxic exposures, nutritional deficiencies, emotional stress, and the complex

impact of social influences. In non-infectious diseases (except for genetic abnormalities), there is usually no factor known to be present in every single case of the disease. For example, smoking is not necessary for the development of lung cancer, and no degree of coronary atherosclerosis is a necessary condition for myocardial infarction.

The *causal inference*, in other words the process of drawing conclusions related to the cause(s) of a disease, is a particularly complex issue in epidemiologic research. Four decades ago, Hill (1971) formalized the criteria that have to be fulfilled in order to accept a causal relation. These included:

1. *Strength of the association.* The stronger the association between the potential (*putative*) risk factor and disease presence, the more likely it is that the anticipated causal relation is valid.
2. *Dose-response effect.* An observation that the frequency of the disease increases with the dose or level of exposure to a certain factor supports a causal interpretation.
3. *Temporal consistency.* It is important to establish that the exposure to the anticipated causative factor occurred prior to the onset of the disease. This may be difficult in cases of diseases with long latent periods or factors that change over time.
4. *Consistency of the findings.* If several studies investigating a given relationship generate similar results, the causal interpretation is strengthened.
5. *Biologic plausibility.* The anticipated relationship should make sense in the context of current biologic knowledge. However, it must be realized that the less that is known about the etiology of a given disease, the more difficult it becomes to satisfy this particular criterion.
6. *Specificity of the association.* If the factor under investigation is found to be associated with only one disease, or if the disease is found to be associated with only one factor among a multitude of factors tested, the causal relation is strengthened. However, this criterion can by no means be used to reject a causal relation, since many factors have multiple effects and most diseases have multiple causes.

It is important to realize that the criteria described above are meant as guidelines for the establishment of a causal inference. None of them, however, is either necessary or sufficient for a causal interpretation. Strict adherence to any of them without concomitant consideration of the others may result in incorrect conclusions.

A distinction has to be drawn between a *causal* factor, assessed as above, and a *risk* factor. In a broad sense, the term risk factor may indicate an aspect of personal behavior or lifestyle, an environmental exposure, or an inborn or inherited characteristic that is known to be associated with disease-related conditions, based on epidemiologic evidence. Such an

attribute or exposure may be associated with an increased probability of occurrence of a particular disease without necessarily being a causal factor. A risk factor may be modified by intervention, thereby reducing the likelihood that the particular disease will occur.

The principles of the *risk assessment process* were discussed by Beck (1994) and consists of the following four steps:

1. The *identification* of one or several individual factors that appear to be associated with the disease.
2. In case of multiple factors, a *multivariate risk assessment model* must be developed that discloses which combination of factors most effectively discriminates between health and disease.
3. The *assessment* step, in which new populations are screened for this particular combination of factors, with a subsequent comparison of the level of the disease assessed with the one predicted by the model.
4. The *targeting* step, in which exposure to the identified factors is reduced by prevention or intervention and the effectiveness of the approach in suppressing the *incidence* of the disease is evaluated.

Thus, according to this process, *potential* or *putative risk factors* (often also referred to as *risk indicators*) are first identified and thereafter tested until their significance as *true risk factors* is proven or rejected.

Finally, distinction must be made between *prognostic* factors (or *disease predictors*), that is characteristics related to the progression of *pre-existing* disease, and *true risk factors*, that is exposures related to the *onset* of the disease. For example, it is established in longitudinal studies of periodontal disease progression (Papapanou *et al.* 1989), that the amount of alveolar bone loss or the number of teeth present at baseline may be used to predict further progression of the disease. These variables are, in fact, alternative measures of the disease itself and express the level of susceptibility of a given subject to periodontal disease. Although they may be excellent predictors for further disease progression, they clearly cannot be considered as risk factors.

There are several ways to study the relation between exposure to a certain factor and the occurrence of a particular disease, as required under step 1. One of these is described in Fig. 7-3, which illustrates a hypothetical situation where exposure to the potential risk factor Z is studied in a cross-sectional study of 1000 subjects, 180 of whom are found to suffer from the disease D ("diseased"), while 820 are disease-free ("healthy"). In this particular setting, it was observed that 155 of the 180 diseased subjects had been exposed to factor Z, but this was also the case for 340 non-diseased subjects. In this example, the association between exposure and disease is expressed by the *odds ratio* (OR), which is the ratio of

	Exposed	Non-exposed	
Diseased	155	25	180
Healthy	340	480	820
	495	505	1000

Fig. 7-3 Contingency table describing the distribution of a group of 1000 subjects according to exposure to a particular factor and disease status.

exposure among the diseased and the healthy. For the data in Fig. 7-3, the OR is calculated as $(155/25)$ over $(340/480) = (155 \times 480) / (340 \times 25) = 8.75$. This indicates that diseased individuals were 8.75 times more likely to be exposed to factor Z than healthy individuals. Note that the OR is frequently misinterpreted to describe the risk of an exposed subject developing the disease, something that is only correctly assessed in a prospective cohort rather than a cross-sectional study and is described by the *relative risk*. In the example of Fig. 7-3, if a sample of 495 subjects exposed to factor Z and 505 subjects not exposed to factor Z are prospectively followed over a given time period, and 155 among the exposed and 25 among the non-exposed develop disease D over this period, then the relative risk is calculated as $(155/495)$ over $(25/505) = 6.4$. In other words, an individual exposed to factor Z is 6.4 times more likely to develop disease D than a non-exposed subject.

In a study of the association between an exposure and the occurrence of a disease, *confounding* can occur when an additional factor is associated with both the exposure and the disease under study. For instance, when examining the association between alcohol consumption and cancer, smoking may act as a confounder, because smokers are frequently alcohol consumers and smoking also has carcinogenic effects.

There are various ways to assess simultaneously the effect of the several putative risk factors identified in step 1 and to generate the *multivariate model* required for step 2. For example, the association between exposure and disease may, for reasons of simplicity, have the form of the following linear equation:

$$y = a - b_1x_1 + b_2x_2 + b_3x_3 + \dots b_nx_n$$

where y represents occurrence or severity of the disease, a is the intercept (a constant value), $x_1, x_2, \dots, x_n$ describe the different exposures (putative risk factors), and $b_1, b_2, \dots, b_n$ are *estimates* defining the

relative importance of each individual exposure as a determinant of disease, after taking all other factors into account. Such an approach will help to identify factors with statistically and biologically significant effects and may minimize the effect of confounders.

In the third step (*assessment step*), a new population sample that is independent of the one used in the construction of the multivariate model is screened for occurrence of disease and presence of the relevant factors included in the multivariate model of step 2. Alternatively, in the case of a prospective cohort study, exposure to the relevant factors is assessed among the subjects of the new sample, and disease incidence, that is the number of new cases of disease, is determined over a time period after a longitudinal follow-up of the subjects. Subsequently, disease occurrence predicted from the model is compared to the actual disease occurrence, and the *external validity* of the model (i.e. the “behavior” or “fitness” of the model in the new population) is evaluated.

Lastly, during the fourth step (*targeting*), aspects of causality or risk are verified if disease occurrence is suppressed when exposure is impeded. Ideally, such studies should be designed as randomized clinical trials, in which treatment is randomly assigned to one of two groups and the effectiveness of the intervention is assessed in direct comparison to outcomes in an untreated, control group. Additionally, an evaluation of the particular preventive/therapeutic strategy from a “cost-benefit” point of view is also facilitated in such studies. Note that successful fulfillment of the targeting step requires that (1) the factor is amenable to intervention and (2) the intervention is delivered at the appropriate time point. Genetic traits are examples of risk factors not amenable to intervention. Likewise, in cases where a single exposure to a factor results in detrimental and/or irreversible biologic damage (e.g. exposure to a high dose of radiation), interventions protecting against subsequent exposure to the factor (radiation) may not lower the incidence of disease (e.g. cancer).

In the context of periodontitis, it should be realized that few of the putative risk factors for disease development have been subjected to the scrutiny of all four steps. In fact, risk assessment studies in dental research in general have been frequently confined to the first two steps. Numerous cross-sectional studies identifying potential risk factors are available, but a relatively limited number of longitudinal studies involve a multivariate approach to the identification of exposures of interest while simultaneously controlling for the effect of possible confounders. Intervention studies in the form of randomized clinical trials are sparse. In the following text, the issue of risk factors is addressed according to the principles described above. Results from cross-sectional studies are considered to provide evidence for putative risk factors that may be further enhanced if corroborated by longitudinal studies involving multivariate techniques, or prospective intervention studies. As reviewed by Borrell and Papapanou (2005), distinction is also made between putative factors that are not amenable to intervention (non-modifiable background factors) and modifiable factors (environmental, acquired, and behavioral).

Non-modifiable background factors

Age

The relationship between age and periodontitis is complex. Although it is clear that both the prevalence and the severity of periodontitis increase with age (Burt 1994; Albandar *et al.* 1999; Dye *et al.* 2007), the concept of periodontitis as an inevitable consequence of aging has been challenged over the years (Papapanou *et al.* 1991; Papapanou & Lindhe 1992) and the alleged “age effect” largely represents the cumulative effect of prolonged exposure to true risk factors. Notably, the association between age and periodontitis appears to be different for pocket depth and amount of clinical attachment loss. While there is a pronounced effect of increasing attachment loss with age, the effect on pocket depth appears to be minimal (Albandar 2002a, b). Interestingly, the effect of age on attachment loss was found to be attenuated after adjustment for co-variables, such as oral hygiene levels or access to dental care services (Albandar, 2002a). In addition, epidemiologic studies have often failed to adjust for important co-variables such as presence of systemic diseases, consumption of multiple medications, and co-morbidities related to nutritional disturbances in the older population, all of which could partly account for the increased prevalence and severity of periodontitis in the elderly. On the other hand, age-associated molecular alterations in key phagocytic cells involved in both the protective and destructive immune responses have been shown to affect their ability to carry out efficient antimicrobial functions and to result in a dysregulation of the inflammatory response (Hajishengallis 2010). Since periodontitis is a

microbially-induced inflammatory disorder, these alterations in innate immunity likely contribute to more pronounced periodontal pathology in elderly individuals. An age-related, rather than an age-dependent, increased susceptibility to periodontitis in older people is therefore biologically plausible.

Sex

There is no established, inherent difference between men and women in their susceptibility to periodontal disease, although men have been shown to exhibit worse periodontal conditions than women in multiple studies from different populations (Brown *et al.* 1989; Albandar 2002a; Susin *et al.* 2004a; Dye *et al.* 2007). This difference has been traditionally considered to reflect the documented better oral hygiene practices (Hugoson *et al.* 1998b; Christensen *et al.* 2003) and/or increased utilization of oral health care services among women (Yu *et al.* 2001; Dunlop *et al.* 2002; Roberts-Thomson & Stewart, 2003). On the other hand, there is evidence for sexual dimorphism in elements of both the innate and the acquired immunity that may lead to enhanced pro-inflammatory responses in men (Shiau & Reynolds 2010), which are in line with the epidemiologic evidence of gender-associated disparities in prevalence, extent, and severity of periodontitis.

Race/ethnicity

Differences in the prevalence of periodontitis between countries and across continents have been demonstrated (Baelum *et al.* 1996; Albandar, 2002a; Dye 2012), but no consistent patterns across racial/ethnic groups have been documented when co-variables such as age and oral hygiene were accounted for (Burt & Eklund 1999). National surveys in the US consistently show a racial/ethnic differential pattern in the prevalence of periodontitis, with African-Americans exhibiting the highest prevalence followed by Mexican Americans and non-Hispanic Caucasians; these findings are fairly consistent regardless of the case definition used (Albandar *et al.* 1999; Arbes *et al.* 2001; Borrell *et al.* 2002; Hyman & Reid 2003; Dye *et al.* 2007; Borrell & Crawford 2008). However, race/ethnicity is usually a social construct that determines an array of opportunities related to access, status, and resources (Williams 1997, 1999). As a result, race/ethnicity and socioeconomic status (SES) are strongly intertwined, suggesting that the observed racial/ethnic effect may be partially attributed to confounding by SES due to the unequal meaning of SES indicators across racial/ethnic groups (Williams 1996; Kaufman *et al.* 1997; Krieger *et al.* 1997; Lynch & Kaplan 2000). Corroborating this point, a study reported that African-Americans demonstrated a lower benefit from education and income in terms of periodontal health status than their Mexican American and Caucasian peers (Borrell *et al.* 2004). These findings confirm that socioeconomic

indicators across racial/ethnic groups are not commensurable but, probably, reflect the broad implications of historical unequal opportunities among certain racial groups (Borrell & Crawford 2012).

Gene polymorphisms

Evidence that genetic predispositions are significant determinants of periodontitis phenotype was first documented in a number of classical twin (Michalowicz *et al.* 1991) and family studies (Boughman *et al.* 1992; Marazita *et al.* 1994). The association of *single nucleotide polymorphisms* (SNPs), that is specific variations at defined locations of the genome that occur in at least 1% of the population, with different forms of periodontitis has been studied extensively (Kinane & Hart 2003; Laine *et al.* 2012). Following the report by Kornman *et al.* (1997) of an association between a composite genotype based on specific polymorphisms in the interleukin-1 (*IL-1*) gene cluster and severe periodontitis in non-smokers, there has been an exponential increase in publications examining a plethora of gene polymorphisms as severity markers of periodontitis. These include additional investigations of the particular composite *IL-1* gene polymorphism in cross-sectional and case-control studies (e.g. Diehl *et al.* 1999; Armitage *et al.* 2000; Papapanou *et al.* 2001; Meisel *et al.* 2004; Li *et al.* 2004), longitudinal studies (Ehmke *et al.* 1999; De Sanctis & Zucchelli 2000; Lang *et al.* 2000; Cullinan *et al.* 2001; Christgau *et al.* 2003; Jepsen *et al.* 2003), as well as studies in which polymorphisms in particular loci of the *IL1A* gene (Fiebig *et al.* 2008; Struch *et al.* 2008), the *IL1B* gene (Lopez *et al.* 2005; Ferreira *et al.* 2008), and the *IL-1* receptor antagonist (Tai *et al.* 2002; Berdeli *et al.* 2006; Fiebig *et al.* 2008) were studied independently of each other.

Similar work has investigated polymorphisms in additional inflammatory genes, including the tumor necrosis factor (*TNF*) gene (Endo *et al.* 2001; Shapira *et al.* 2001; Craandijk *et al.* 2002; Fassmann *et al.* 2003; Shimada *et al.* 2004); the *IL-6* gene (Anusaksathien *et al.* 2003; Holla *et al.* 2004; Nibali *et al.* 2009); the *IL-4* gene (Kang *et al.* 2003; Holla *et al.* 2008; Kobayashi *et al.* 2009); and the *IL-10* gene (Brett *et al.* 2005; Kobayashi *et al.* 2009). A substantial body of data has accumulated on polymorphisms in genes coding for various receptors, including the leukocyte receptors for the constant part (Fc) of immunoglobulin G (Kobayashi *et al.* 1997; Sugita *et al.* 1999; Kobayashi *et al.* 2000a, b, 2001; Meisel *et al.* 2001; Loos *et al.* 2003; Yamamoto *et al.* 2004; Wolf *et al.* 2006; Nibali *et al.* 2006); pattern recognition receptors such as CD14 (Holla *et al.* 2002; James *et al.* 2007; Tervonen *et al.* 2007) and Toll-like receptors (TLRs) 2 and 4 (Folwaczny *et al.* 2004; Brett *et al.* 2005; Fukusaki *et al.* 2007; Noack *et al.* 2008); and the vitamin D receptor (Park *et al.* 2006; Nibali *et al.* 2008; Wang *et al.* 2009). Additional polymorphisms studied in single studies or a few cohorts have been discussed in a recent comprehensive review (Laine *et al.* 2012) and a meta-analysis of

53 studies collectively including 4178 cases and 4590 controls (Nikolopoulos *et al.* 2008).

Typically, the majority of these cross-sectional studies report positive associations between the investigated polymorphisms and the extent or the severity of periodontitis. The results, however, are not unequivocal, as the strength of the reported associations is not uniformly consistent across populations, the frequency of occurrence of these polymorphisms appears to vary extensively between ethnic groups, the subject samples involved are generally of limited size, the definitions of the outcome variable (periodontitis) vary considerably, and adequate adjustments for other important co-variables and risk factors have frequently not been carried out. Importantly, there appear to be differences in the impact of these polymorphisms on early-onset versus adult forms of periodontitis. For example, in the case of *IL-1* polymorphisms, while it is the rare allele (allele 2) that has been linked with severe disease in adults, it is allele 1 that has been found to be more prevalent in subjects with early-onset periodontitis (Diehl *et al.* 1999; Parkhill *et al.* 2000).

The results of the relatively few longitudinal studies that have studied specific gene polymorphisms as exposures are also conflicting. Ehmke *et al.* (1999) reported no bearing of the *IL-1* gene polymorphism on the prognosis of periodontal disease progression following non-surgical periodontal therapy. Jepsen *et al.* (2003) failed to provide evidence that the *IL-1* risk genotype was associated with higher gingival crevicular fluid (GCF) volume and percentage of bleeding on probing (BoP) sites during the development of experimental gingivitis (Löe *et al.* 1965). In contrast, Lang *et al.* (2000) concluded that *IL-1* genotype-positive subjects have a genetically determined hyper-inflammatory response that is expressed clinically in the periodontal tissues as increased prevalence and incidence of BoP during maintenance. Three treatment studies examined the impact of this particular polymorphism in regenerative therapy: De Sanctis and Zucchelli (2000) reported that the *IL-1*-positive genotype was associated with an inferior long-term outcome for regenerative therapy of intra-bony defects. In contrast, Christgau *et al.* (2003) and Weiss *et al.* (2004) failed to document such an association in similar studies. Finally, in a 5-year prospective study of 295 subjects, Cullinan *et al.* (2001) reported an interaction between the positive genotype, age, smoking, and colonization by *Porphyromonas gingivalis*, concluding that the positive genotype is a contributory but non-essential factor for the progression of periodontal disease. A systematic review of longitudinal studies that examined the composite *IL-1* genotype as a predictor of periodontitis progression or of periodontal treatment outcomes failed to establish a significant association (Huynh-Ba *et al.* 2007).

In conclusion, there is insufficient epidemiologic evidence that convincingly establishes any of the above polymorphisms as true risk factors for

periodontitis. Studies employing larger cohorts, strict classification criteria for periodontitis, and refined analytical methods will enhance our understanding of the role of genetic influences in the pathobiology of periodontitis.

Environmental, acquired, and behavioral factors

Specific microbiota

The microbial etiology of experimental gingivitis (Löe *et al.* 1965; Theilade *et al.* 1966) and periodontitis (Lindhe *et al.* 1973) has been established for several decades. Yet, epidemiologic studies that systematically investigated the role of specific microbiota as putative risk factors for periodontitis were undertaken only fairly recently. In a classic paper, Haffajee and Socransky (1994) adapted Koch's postulates for use in the identification of periodontal pathogens and proposed the following criteria:

1. Association, that is elevated ORs in disease
2. Elimination, that is conversion of periodontal disease to health when bacteria are eliminated (or suppressed beyond detection)
3. Development of a host response, typically manifested by a serum antibody titer to the infecting agent
4. Presence of virulence factors that can account for the microbe's ability to inflict tissue damage
5. Evidence from animal studies that corroborate the observations in humans and demonstrate development of periodontal pathology after infection by the microorganism.

Based on these criteria, the consensus report of the 1996 World Workshop in Periodontics identified three species, *Actinobacillus actinomycetemcomitans*, *P. gingivalis*, and *Bacteroides forsythus*, as causative factors for periodontitis [since then, two of these three causative species have been renamed: *A. actinomycetemcomitans* to *Aggregatibacter actinomycetemcomitans* (Norskov-Lauritsen & Kilian 2006) and *B. forsythus* to *Tannerella forsythia* (Sakamoto *et al.* 2002; Maiden *et al.* 2003)]. However, given that only approximately 50% of the bacteria of the oral cavity are currently recognized (Paster *et al.* 2001), it is clear that these three species cannot be considered to be the only causative pathogens, but are rather the ones for which sufficient data have accumulated.

Over the last decade, interesting data have emerged on the prevalence of these causative bacteria in different populations, including periodontally healthy as well as diseased subjects. Studies performed in children (Yang *et al.* 2002; Tanner *et al.* 2002) analyzed plaque from the gingival crevice, tooth surface, and dorsum of the tongue, and revealed that sizeable proportions of subjects harbored *P. gingivalis*, *T. forsythia*, and *A. actinomycetemcomitans* despite

absence of overt gingival inflammation. A comparably high carrier state was documented in studies that sampled infants, children, adolescents, and adults with apparently healthy periodontal conditions (Könönen 1993; McClellan *et al.* 1996; Kamma *et al.* 2000; Lamell *et al.* 2000). Thus, contrary to the conclusions of earlier, culture-based studies suggesting that these bacteria occur infrequently in periodontally healthy oral cavities and behave as exogenous pathogens, use of contemporary molecular techniques for bacterial identification have demonstrated that this is not the case. However, both the prevalence and the level of colonization by these pathogens have been shown to vary significantly between populations of different racial or geographic origin (Ali *et al.* 1994; Sanz *et al.* 2000; Lopez *et al.* 2004; Haffajee *et al.* 2004; Rylev & Kilian 2008).

Several epidemiologic studies have examined the prevalence of the established periodontal pathogens and its relation to clinical periodontal status in population samples from both developed and developing countries. Griffen *et al.* (1998) examined a convenience sample recruited from a university clinic, and reported that 79% of the diseased and 25% of the healthy subjects were positive for *P. gingivalis*. Interestingly, the prevalence of *P. gingivalis* in the periodontally healthy group varied substantially with race/ethnicity, occurring in 22% of Caucasians, 53% of African-Americans, and 60% of Asian-Americans. In a case-control study of periodontitis patients and age- and sex-matched controls with no or only minimal attachment loss in Sweden, Papapanou *et al.* (2000) reported a high prevalence of *P. gingivalis*, *A. actinomycetemcomitans*, *T. forsythia*, and *Treponema denticola* in periodontitis patients (95%, 83%, 97%, and 93%, respectively), but also similarly high prevalence rates among control subjects (82%, 90%, 82%, and 94%, respectively). However, in a quantitative analysis of bacterial load, substantial differences in colonization at high levels (i.e. at an average count of $\geq 10^5$ bacterial cells/plaque sample) were observed between patients and controls for three of the four bacteria: 19% versus 3% for *P. gingivalis*; 54% versus 12% for *T. forsythia*, and 46% versus 19% for *T. denticola*. In contrast, corresponding percentages were similar for *A. actinomycetemcomitans* (1% in both cases and controls). Substantially different prevalence data were reported in a study of blue- and white-collar university employees in Australia (Hamlet *et al.* 2001). These authors detected *A. actinomycetemcomitans* in 23% and *P. gingivalis* in 15% of the subjects.

A number of studies investigated the epidemiology of periodontal pathogens in Asian populations. Timmerman *et al.* (1998) examined a sample of adolescents in rural Indonesia, and detected *P. gingivalis* in 87% and *A. actinomycetemcomitans* in 57%. Mombelli *et al.* (1998) examined young factory workers in China and detected *A. actinomycetemcomitans* in 62% and *P. gingivalis* in 55%. In contrast, an almost ubiquitous presence of *P. gingivalis* and *T. forsythia* was reported

in rural subject samples in China (Papapanou *et al.* 1997) and Thailand (Papapanou *et al.* 2002), while *A. actinomycetemcomitans* was detected in 83% and 93% of the subjects in the Chinese and Thai samples, respectively. Despite this high prevalence, a quantitative analysis of bacterial load correlated well with periodontal status in both studies. For example, a discriminant analysis performed on the data from the Thai study (Papapanou *et al.* 2002) identified threshold levels of average bacterial load which, when exceeded, conferred increased ORs for the presence of three or more sites with a pocket depth of ≥ 5 mm. For three species [*P. gingivalis*, *T. forsythia*, and *T. denticola*; the “red complex” bacteria (Socransky *et al.* 1998)], colonization above these calculated thresholds resulted in statistically significant, elevated ORs for periodontitis. In addition, an analysis of the association between colonization at high levels by the “red complex” bacteria and specific periodontal conditions, defined in this particular study by the presence of three or more sites with a pocket depth of ≥ 5 mm and by two different extents of clinical attachment loss (≥ 10 and ≥ 30 sites with ≥ 5 mm attachment loss, respectively), revealed statistically significant ORs ranging between 3.7 and 4.3 for the “red complex” bacteria and all three disease definitions. Similar cross-sectional associations of statistically significant ORs for severe periodontitis conferred by specific bacteria have been also observed in several other studies involving subject samples from the Western world (Grossi *et al.* 1994, 1995; Alpagot *et al.* 1996; Craig *et al.* 2001).

Importantly, the association between high levels of colonization by specific periodontal pathogens and the progression of periodontal disease has been corroborated by longitudinal data in untreated populations. For example, in the study by Papapanou *et al.* (1997), a discriminant analysis based on quantitative assessments of subgingival bacterial load classified correctly the substantial majority of the subjects with progression of periodontitis over a preceding 10-year period. Indeed, bacterial profiles classified correctly 75% of the subjects with ten or more sites with longitudinal attachment loss of ≥ 3 mm, and 85% of those that remained stable over the observation period. In a 7-year follow-up study of Indonesian adolescents (Timmerman *et al.* 2000, 2001), and in a subsequent 15-year follow-up of the same cohort (van der Velden *et al.* 2006), it was shown that the subgingival presence of *A. actinomycetemcomitans* was associated with disease progression, defined as presence of longitudinal attachment loss of ≥ 2 mm. In a follow-up of 2–5 years’ duration, Machtei *et al.* (1999) reported that subjects colonized by *T. forsythia* at baseline exhibited greater alveolar bone loss, a larger proportion of “loser” sites, that is sites exhibiting progressive attachment loss, and twice as high longitudinal tooth loss than non-colonized subjects. In a 3-year study, Hamlet *et al.* (2004) reported ORs of 8.2 for attachment loss in adolescents with persistent colonization with *T. forsythia*.

An important observation was made in a 2-year prospective study of clinical periodontal status in adolescents in Morocco (Haubek *et al.* 2008). These investigators reported that colonization by a specific clone of *A. actinomycetemcomitans*, namely the highly-leukotoxic JP2 clone, conferred a much higher risk for the onset of aggressive periodontitis in periodontally healthy schoolchildren than concomitant colonization by a variety of clones of the same species, or the total absence of colonization by *A. actinomycetemcomitans*. Indeed, in comparison to schoolchildren who were not colonized by *A. actinomycetemcomitans*, the relative risk for incident disease in those colonized exclusively by JP2 clones was 18.0 (95% CI 7.8–41.2), as compared to 12.4 (95% CI 5.2–29.9) in those colonized by both JP2 and non-JP2 clones, and 3.0 (95% CI 1.3–7.1) in those colonized exclusively by non-JP2 clones of *A. actinomycetemcomitans*. This study underscored the important role of this particular periodontal pathogen in the etiology of aggressive periodontitis, but also demonstrated that within-species variation in virulence is associated with differences in the clinical presentation of the disease.

Collectively, data generated in the past 20 years have enhanced our knowledge of the role of specific periodontal bacteria as risk factors for periodontitis (Table 7-4), and have clarified that (1) the intensity of the exposure to the specific microbiota (“pathogen burden”) rather than the mere presence of the pathogen is an important determinant of the clinical phenotype; (2) the virulence of the pathogen, and thus its ability to cause periodontal tissue damage and confer risk for disease progression may be entirely different among various clonal types within a single species; and (3) pathogen elimination from the subgingival microbiota (or more correctly, suppression of its levels beyond detection) results in improvements in clinical periodontal status. This last “targeting” criterion of the risk assessment process described earlier has been abundantly fulfilled in the case of microbial risk factors. As demonstrated in systematic reviews, an antimicrobial approach, including removal of subgingival plaque with or without adjunctive antiseptics or antibiotics followed by adequate maintenance care, is the single most successful and consistent strategy in the treatment of periodontitis (Herrera *et al.* 2002; Heitz-Mayfield *et al.* 2002; Tonetti & Chapple 2011).

Cigarette smoking

The biologic plausibility of an association between tobacco smoking and periodontitis was founded on the broad effects of multiple tobacco-related substances on cellular structure and function. Smoking has been shown to affect the vasculature, the humoral and cellular immune responses, cell signaling processes, and tissue homeostasis [for review see Kinane and Chestnutt (2000) and Palmer *et al.* (2005)]. A substantial number of studies, a selection of which is

Table 7-4 Selected studies using bacteria as exposures of significance for periodontitis.

Authors/country ^a	Sample/methodology	Findings
Beck <i>et al.</i> (1990) USA	690 community dwelling adults, aged 65+ years; probing assessments at mesio- and mid-buccal surfaces, all teeth; logistic regression for advanced AL and deep pocketing; "advanced disease": ≥ 4 sites with AL of ≥ 5 mm and ≥ 1 of these sites with PD of ≥ 4 mm	Blacks: 78% of sites with attachment loss, mean AL on these sites 4 mm Caucasians: 65%, 3.1 mm OR in Blacks: tobacco use 2.9; <i>Porphyromonas gingivalis</i> $>2\%$ 2.4; <i>Prevotella intermedia</i> $>2\%$ 1.9; last dental visit >3 years 2.3; bleeding gums 3.9 OR in Caucasians: tobacco use 6.2; presence of <i>P. gingivalis</i> (+) 2.4; no dental visits for >3 years plus BANA (+) 16.8
Haffajee <i>et al.</i> (1991b) USA	38 subjects, aged 14–71 years, with prior evidence of attachment loss; 2-month follow-up; probing assessments at six sites/tooth, all teeth; 28 subgingival samples per subject at baseline, DNA probe analysis with respect to 14 bacterial species; progression threshold: ≥ 3 mm of longitudinal AL; the mean percentage of the total cultivable microbiota was averaged across active and inactive sites; OR computed at different thresholds for each species	Significant odds ratios for new disease: <i>P. gingivalis</i> 5.6, <i>Campylobacter rectus</i> 3.8, <i>Veillonella parvula</i> 0.16 and <i>Capnocytophaga ochracea</i> 0.08 Discriminant analysis using the significantly related species was useful in predicting subjects at risk for new attachment loss
Grossi <i>et al.</i> (1994) USA	Random sample of 1426 subjects, aged 25–74 years, in a metropolitan community; full-mouth probing assessments; multivariate analysis of risk indicators for attachment loss. Exposures: (1) clinical – supragingival plaque, gingival bleeding, subgingival calculus, PD, CAL; (2) microbial – <i>A. actinomycetemcomitans</i> , <i>T. forsythia</i> , <i>C. rectus</i> , <i>Eubacterium saburreum</i> , <i>Fusobacterium nucleatum</i> , <i>P. gingivalis</i> , <i>Capnocytophaga</i> spp and <i>P. intermedia</i> ; (3) co-variables – age, gender, race, education, income, smoking and numbers of packs/year, exposure to occupational hazards, systemic diseases	In a multivariable logistic regression model, <i>P. gingivalis</i> (OR = 1.59; 95% CI 1.11–2.25) and <i>T. forsythia</i> (OR = 2.45; 95% CI 1.87–3.24) were positively associated with severity of AL, while <i>Capnocytophaga</i> spp. (OR = 0.60; 95% CI 0.43–0.84) were protective against AL
Grossi <i>et al.</i> (1995) USA	Same sample as in Grossi <i>et al.</i> (1994); 1361 subjects, aged 25–74 years; assessments of interproximal bone loss from full-mouth radiographs; the degree of association between bone loss and explanatory variables was analyzed by stepwise logistic regression	In a multivariable logistic regression model, <i>P. gingivalis</i> (OR = 1.73; 95% CI 1.27–2.37) and <i>T. forsythia</i> (OR = 2.52; 95% CI 1.98–3.17) were significantly associated with increasing severity of bone loss
Beck <i>et al.</i> (1997) USA (L)	540 dentate adults, aged 65+ years, examined at baseline, 18, 36 and 60 months; incidence of AL was defined as additional AL ≥ 3 mm; microbial variables included presence of <i>A. actinomycetemcomitans</i> , <i>P. intermedia</i> , and <i>P. gingivalis</i> , and the BANA test; co-variables included age, gender, missing teeth, education, smoking, dental visits	BANA (+), and presence of <i>P. gingivalis</i> were significantly associated with incident disease
Papapanou <i>et al.</i> (1997) China (L)	148 subjects, aged 30–39 and 50–59 years in a rural area examined 10 years apart; Full-mouth assessments of PD and AL at six sites/tooth; 14 subgingival plaque samples were obtained from each subject at the follow-up examination (1864 in total) and analyzed with respect to 18 bacterial species	Ubiquitous prevalence for the majority of the investigated species at the subject level. Bacterial colonization at high levels by <i>P. gingivalis</i> , <i>P. intermedia</i> , <i>Prevotella nigrescens</i> , <i>T. forsythia</i> , <i>F. nucleatum</i> , <i>Treponema denticola</i> , <i>Parvimonas micra</i> , and <i>C. rectus</i> conferred statistically significant OR for being classified as "downhill" (≥ 10 sites with longitudinal AL loss of ≥ 3 mm)
Machtei <i>et al.</i> (1999) USA (L)	A sample of 415 subjects, aged 25–75 years, followed for a period of 2–4 years; full-mouth examination at six sites/tooth at all teeth present; full-mouth intraoral radiographs; bacterial samples obtained from 12 index teeth analyzed with respect to <i>A. actinomycetemcomitans</i> , <i>T. forsythia</i> , <i>C. rectus</i> , <i>P. intermedia</i> , <i>Capnocytophaga</i> spp, <i>P. gingivalis</i> , <i>E. saburreum</i> , and <i>F. nucleatum</i> ; co-variables included age, gender, smoking (current smokers 15.4%), education, income	Subjects harboring <i>T. forsythia</i> at baseline showed significantly higher longitudinal bone loss, greater proportion of "loser sites" (sites with additional AL of ≥ 2 mm) and twice as high tooth mortality

(Continued)

Table 7-4 Continued

Authors/country ^a	Sample/methodology	Findings
Timmerman <i>et al.</i> (2000) Indonesia (L)	A sample of 255 subjects, aged 15–25 years, in a rural area, examined 7 years apart; assessments of PD and AL at vestibular surfaces of all teeth; bacterial samples harvested from a variety of intraoral sites and analyzed with respect to <i>A. actinomycetemcomitans</i> , <i>P. gingivalis</i> , <i>P. intermedia</i> , spirochetes and motile microorganisms	Progressive disease (PDS) was defined as ≥ 1 site with longitudinal AL ≥ 2 mm Subgingival presence of <i>A. actinomycetemcomitans</i> (OR 4.2; 95% CI: 1.4–12.7), <i>P. gingivalis</i> (OR 2.3; 95% CI 1.0–5.2), and motile microorganisms (OR 2.2; 95% CI 1.0–5.0) were associated with PDS In a multivariable logistic model, including age and subgingival calculus, subgingival presence of <i>A. actinomycetemcomitans</i> (OR=4.61, $P=0.01$) was associated with PDS
Papapanou <i>et al.</i> (2002) Thailand	Random sample of 356 subjects aged 30–39 and 50–59 years, in a rural area; PD and CAL were assessed at six sites/tooth, at all teeth apart from third molars; subjects were grouped according to different levels of pocketing/attachment loss: subjects with ≥ 3 sites with PD ≥ 5 mm (59%, G1); ≥ 10 sites with CAL ≥ 5 mm (50%, G2); and ≥ 30 sites with CAL ≥ 5 mm (24%, G3) Subgingival plaque samples were obtained maximally at 14 sites/subject; checkerboard hybridizations were used to analyze a total of 4343 samples with respect to 27 bacterial species	OR for heavy colonization by “red complex” species (<i>P. gingivalis</i> , <i>T. forsythia</i> , <i>T. denticola</i>) were 3.7 (95% CI 2.3–5.9) for G1; 4.0 (95% CI 2.5–6.6) for G2; and 4.3 (95% CI 2.6–7.1) for G3 OR for heavy colonization by selected “orange complex” species (<i>F. nucleatum</i> , <i>P. intermedia</i> , <i>P. nigrescens</i> , <i>P. micros</i> , <i>E. nodatum</i> , <i>C. rectus</i> , and <i>Campylobacter showae</i>) were 1.5 (95% CI 0.8–2.9) for G1; 1.5 (95% CI 0.8–2.9) for G2; and 1.5 (95% CI 0.8–3.1) for G3
van der Velden <i>et al.</i> (2006) Indonesia (L)	15-year follow-up of 128 subjects from the cohort originally described by Timmerman <i>et al.</i> (2000)	In a multivariable logistic model, subgingival presence of <i>A. actinomycetemcomitans</i> (OR 4.3; 95% CI 1.2–15.7) was confirmed as a risk factor for the onset of the disease, i.e. longitudinal AL during the first 7-year period, but not for progression of disease during the subsequent 8-year period
Fine <i>et al.</i> (2007) USA (L)	1075 primarily African-American and Hispanic children aged 11–17 years were examined periodontally and with respect to oral colonization by <i>A. actinomycetemcomitans</i> ; 96 children were followed up at 6-month intervals for at least 2.5 years; 38 were <i>A. actinomycetemcomitans</i> -positive at baseline (36 periodontally healthy and 2 with pathologic periodontal pockets) and 58 were <i>A. actinomycetemcomitans</i> -negative and periodontally healthy at baseline	In the cross-sectional cohort, 4% of the children had signs of periodontal pathology (≥ 1 pocket with ≥ 6 mm PD and > 2 mm AL; 67% of those were <i>A. actinomycetemcomitans</i> -positive, as compared with 14% of the entire cohort In the longitudinal cohort, 21% of the <i>A. actinomycetemcomitans</i> -positive and 0% of the <i>A. actinomycetemcomitans</i> -negative children developed bone loss over the follow-up period
Haubek <i>et al.</i> (2008) Morocco (L)	700 adolescents from public schools in Rabat were examined for clinical periodontal status and presence of <i>A. actinomycetemcomitans</i> assessed by PCR; 682 were periodontally healthy; of these, 428 returned for a follow-up exam 2 years later	Individuals who carried the high leukotoxic JP2 clone of <i>A. actinomycetemcomitans</i> alone (RR 18.0; 95% CI 7.8–41.2) or together with non-JP2 clones (RR 12.4; 95% CI 5.2–29.9) had a significantly increased risk of developing periodontitis Risk for those carrying non-JP2 clones alone was less pronounced (RR 3.0; 95% CI 1.3–7.1)

^aL indicates a longitudinal study.

PD, probing depth; AL, attachment level; CAL, clinical attachment level; CEJ, cemento-enamel junction; CPITN, Community Periodontal Index of Treatment Needs; BANA: N-benzoyl-DL-arginine-2-naphthylamide; a substrate hydrolyzed in the presence of *T. denticola*, *Porphyromonas gingivalis*, and *Tannerella forsythia*; CI, confidence interval; OR, odds ratio; RR, risk ratio; PCR, polymerase chain reaction.

summarized in Table 7-5, established the association of smoking with poor periodontal status. Importantly, the inferior periodontal status of smokers cannot be attributed to poorer plaque control or more severe gingivitis (Bergström 1989). While earlier reports suggested a rather similar composition of the subgingival microflora in smokers and non-smokers (Stoltenberg *et al.* 1993), recent studies demonstrated that smoking contributes to the formation of a dysbiotic biofilm. It

affects bacterial acquisition and colonization (Brook 2011; Kumar *et al.* 2011), bacterial aggregation (Bagaitkar *et al.* 2011), and results in higher levels of colonization by key periodontal pathogens (Haffajee & Socransky 2001; Shchipkova *et al.* 2010; Kubota *et al.* 2011). In an attempt to quantitate the effects of smoking on the periodontal conditions, Haber *et al.* (1993) suggested that the excess prevalence of periodontal disease in the population attributed solely to

Table 7-5 Selected studies using smoking as exposure of significance for periodontitis.

Authors/country ^a	Sample/methodology	Findings
Bergström (1989) Sweden	Patients referred for periodontal therapy (155 subjects, aged 30, 40 and 50 years); a random sample of the Stockholm population served as controls; full-mouth probing assessments; sites with PD $\geq$ 4 mm considered diseased; recording of plaque and gingivitis scores	56% of the patients and 34% of the controls were smokers (OR 2.5); significantly higher frequency of periodontally-involved teeth in smokers; no notable difference between smokers and non-smokers with respect to plaque and gingivitis
Haber & Kent (1992) USA	196 patients with periodontitis in a periodontal practice and 209 patients from five general practices; probing assessments at six sites/tooth and full-mouth radiographs; questionnaire on smoking habits; patients with negative history of periodontal therapy from the general practices included as controls; comparison of (1) the prevalence of smoking among the two patient groups and (2) periodontitis/disease severity among current and never smokers	Overall smoking history in the periodontal practice 75%; in the general practice 54% Summary OR for positive smoking history in periodontal versus general practice patients was 2.6; in the periodontal group, frequency of current smoking increased with increasing severity of periodontitis
Locker (1992) Canada	907 adults, aged $\geq$ 50 years, living independently in four Ontario communities; partial, probing assessments; half of the participants reported a positive history of smoking and 20% were current smokers	Current smokers had fewer teeth, were more likely to have lost all their natural teeth, and had higher extent and severity of periodontitis than those who had never smoked
Haber et al. (1993) USA	132 diabetics and 95 non-diabetics, aged 19–40 years; probing assessments at six sites/tooth, all teeth; questionnaire on smoking habits; calculation of the population attributable risk percent (PAR%), as an estimate of the excess prevalence of periodontitis in the study population that is associated with smoking	Prevalence of periodontitis was markedly higher among smokers than non-smokers within both the diabetic and non-diabetic groups; PAR% among non-diabetics was 51% in those aged 19–30 years and 32% in those aged 31–40 years
Stoltenberg et al. (1993) USA	Of 615 medically healthy adults, aged 28–73 years, attending a health maintenance organization, selection of 63 smokers and 126 non-smokers of similar age, sex, plaque and calculus scores; probing assessments at the proximal surfaces of premolars and molars in a randomly selected posterior sextant; detection of <i>Porphyromonas gingivalis</i> , <i>Prevotella intermedia</i> , <i>Aggregatibacter actinomycetemcomitans</i> , <i>Eubacterium saburreum</i> , and <i>Fusobacterium nucleatum</i> . by a semiquantitative fluorescence immunoassay, in one buccal and one lingual sample per tooth examined; logistic regression to determine if any of the bacteria or smoking were indicators of mean posterior probing depth of $\geq$ 3.5 mm	OR for a smoker having a mean PD of $\geq$ 3.5 mm was 5.3 (95% CI 2.0–13.8) No statistically significant difference between smokers and non-smokers with respect to prevalence of the bacteria examined Logistic model revealed that a mean PD of $\geq$ 3.5 mm was significantly associated with the presence of <i>A. actinomycetemcomitans</i> , <i>P. intermedia</i> , <i>E. saburreum</i> , and smoking; smoking was a stronger indicator than any of the bacteria examined
Jette et al. (1993) USA	1156 community dwellers, aged 70+ years; probing assessments at four sites/tooth, all teeth; evaluation of lifelong tobacco use as a modifiable risk factor for poor dental health; multiple regression analysis	18.1% of men and 7.9% of women were tobacco users (overall 12.3%; including 1% smokeless tobacco users) Years of exposure to tobacco products was a statistically significant factor for tooth loss, coronal root caries, and periodontal disease, regardless of other social and behavioral factors Periodontal disease (number of affected teeth) was predicted by longer duration of tobacco use, male sex, and more infrequent practice of oral hygiene
Martinez Canut et al. (1995) Spain	889 periodontitis patients, aged 21–76 years; probing assessments at six sites/tooth, all teeth; analysis of variance to examine the role of smoking on the severity of periodontitis	Smoking was statistically related to increased severity of periodontitis in multivariate analysis; a dose–response effect was demonstrated, with subjects smoking $>$ 20 cigarettes/day showing significantly higher attachment loss
Kaldahl et al. (1996) USA (L)	74 patients with moderate to advanced periodontitis, including 31 heavy smokers ($\geq$ 20 cigarettes/day); the effects of cigarette consumption and smoking history on the response to active periodontal treatment and to up to 7 years of supportive periodontal treatment was evaluated Full-mouth examinations performed at baseline, 4 weeks after mechanical plaque control, 10 weeks following periodontal surgery, and yearly during 7 years of supportive periodontal treatment	Past and never smokers consistently exhibited a significantly greater reduction in PD and greater gains in AL than heavy and light smokers All groups experienced a similar decrease in the prevalence of BoP following active therapy

(Continued)

Table 7-5 Continued.

Authors/country ^a	Sample/methodology	Findings
Grossi <i>et al.</i> (1997) USA (L)	143 subjects aged 35–65 years with established periodontitis, including 60 current, 55 former, and 28 non-smokers, examined at baseline and 3 months after non-surgical periodontal therapy	Current smokers showed less reduction in PD and less AL gain than former- and non-smokers Fewer smokers harbored no <i>P. gingivalis</i> or <i>Tannerella forsythia</i> after treatment, compared to former and non-smokers
Axelsson <i>et al.</i> (1998) Sweden	A random sample of 1093 subjects, aged 35, 50, 65, and 75 years; prevalence of smoking in the four age groups was 35%, 35%, 24%, and 12%, respectively; recordings included AL, CPITN scores, DMF surfaces, plaque, and stimulated salivary secretion rate (SSSR)	In the oldest age group, 41% of the smokers and 35% of the non-smokers were edentulous In every age group, mean AL was statistically significantly increased in smokers by 0.37, 0.88, 0.85, and 1.33 mm, respectively Smokers had higher CPITN and DMF scores, increased SSSR, but similar plaque levels
Tomar & Asma (2000) USA	12 329 subjects, aged ≥18 years, participants in the NHANES III study; probing assessments at mesial and buccal sites in one upper and one lower quadrant; mesial assessments performed from the buccal aspect of the teeth; assessments of gingivitis, PD, and location of the gingival margin in relation to the CEJ; “periodontitis” was defined as ≥1 site with AL ≥4 mm and PD ≥4 mm	27.9% of participants were current smokers and 9.2% met the definition for periodontitis Current smokers were four times as likely to suffer from periodontitis than never smokers, after adjustments for age, gender, race/ethnicity, education, and income:poverty ratio Among current smokers, there was a dose–response relationship between cigarettes/day and periodontitis 41.9% of periodontitis cases were attributable to current smoking and 10.9% to former smoking
Bergström <i>et al.</i> (2000b) Sweden	257 subjects, aged 20–69 years, including 50 current smokers, 61 former smokers, and 133 non-smokers; full-mouth clinical and radiographic assessments of the periodontal tissues; smoking exposure defined in terms of consumption (number of cigarettes/day), duration (number of years of smoking) and lifetime exposure (product of daily consumption and years of duration – cigarettes/year); threshold levels used: heavy versus light consumption: ≥10 cigarettes/day versus <10 cigarettes/day; duration: ≥15 years versus <15 years; lifetime exposure: ≥200 cigarettes/year versus <200 cigarettes/year	Compared to former and non-smokers, current smokers had the highest prevalence of diseased sites (AL ≥4 mm) 40–69-year-old current smokers showed a significantly higher prevalence than 20–39-year-old current smokers (27% versus 4%) The same pattern emerged when comparing heavy versus light smokers according to consumption, duration, and lifetime exposure; in multiple regression, lifetime exposure was highly associated with the frequency of diseased sites and periodontal bone height after adjusting for age, gingival bleeding, and plaque index
Albandar <i>et al.</i> (2000) USA	705 subjects, aged 21–92 years, 52% males and 87% Caucasian; full-mouth examination of PD and AL at six sites; periodontitis was classified as advanced, moderate, or mild; cigar, pipe, and cigarette smoking were classified as current, former, and never	In multiple linear regression, current and former smoking, regardless of type, was associated with increased percentage of subjects with moderate/advanced periodontitis after adjusting for age, gender, race, and number of years of cigarette, cigar, and pipe smoking; current smoking was also associated with higher number of missing teeth
Bergström <i>et al.</i> (2000a) Sweden (L)	10-year follow-up of a sample of 84 dentally aware musicians, including 16 current, 28 former, and 40 non-smokers; full-mouth clinical and radiographic assessments of periodontal status	Prevalence of PD ≥4 mm (“diseased” sites) was 18.7% for current, 11.1% for former, and 8.7% for non-smokers at baseline. At 10 years, these figures were 41.6%, 7.8%, and 6.6%, respectively; a similar pattern was observed for alveolar bone levels After adjusting for age, gingival bleeding, plaque index, and frequency of diseased sites at baseline, current smoking was a significant predictor of the increase in diseased sites at 10 years
Susin <i>et al.</i> (2004b) Brazil	974 subjects, aged 30–103 years; full-mouth examination of PD and AL; severe attachment loss was defined as AL ≥5 mm in ≥30% of the teeth; exposure to smoking classified as current/former, heavy/moderate/light/none, and quantified as lifetime consumption	Heavy and moderate smokers had significantly higher prevalence of AL ≥5 mm than non-smokers; in multivariate analysis heavy (OR=3.6; 95% CI 2.2–6.0) and moderate smoking (OR=2.0; 95% CI 1.4–2.9) conferred higher OR for AL; the attributable fraction of AL due to smoking was 37.7% and 15.6% among heavy and moderate smokers, respectively

(Continued)

Table 7-5 Continued

Authors/country ^a	Sample/methodology	Findings
Phipps <i>et al.</i> (2009) USA	1210 dentate men aged ≥65 years; random half-mouth assessments of PD and AL at six sites/tooth; "periodontitis" was defined as (1) presence of interproximal attachment loss ≥5 mm in ≥30% of the teeth present (5th European Workshop definition), or (2) according to the CDC/AAP definition; information on smoking habits collected through self-administered questionnaires and converted to pack-years	Prevalence of periodontitis according to the 5th European workshop definition was 30.4% among never smokers, 35.3% among smokers with <20 pack-years, and 50.3% among smokers with ≥20 pack-years ($P < 0.001$); corresponding prevalence figures according to the CDC/AAP definition were 22.0%, 19.2%, and 36.4%, respectively ($P < 0.001$). In multivariate analysis adjusted for age, race, and dental visit attendance, ≥20 pack-years were significantly associated with periodontitis OR 2.11 (95% CI 1.64–2.72) and 2.38 (95% CI 1.79–3.13), according to the two definitions, respectively

^aL indicates a longitudinal study

PD, probing depth; AL, attachment level; BoP, bleeding on probing; CEJ, cemento-enamel junction; CPITN, Community Periodontal Index of Treatment Needs; NHANES, National Health and Nutrition Examination Survey; DMF, decayed, missing, filled; CI, confidence interval, OR, odds ratio.

smoking is much greater than that due to other systemic predispositions, such as diabetes mellitus. Data derived from the NHANES III study (Tomar & Asma 2000) suggested that as many as 42% of periodontitis cases in the US can be attributed to current smoking and another 11% to former smoking. Similarly, a study from Brazil (Susin *et al.* 2004b) reported that the attributable fraction of clinical attachment loss due to cigarette smoking was 37.7% and 15.6% among heavy and moderate smokers, respectively. In longitudinal studies, smoking has been found to confer a statistically significant increased risk for periodontitis progression after adjustment for other co-variables (Beck *et al.* 1995, 1997; Machtei *et al.* 1999; Norderyd *et al.* 1999; Chen *et al.* 2001; Ogawa *et al.* 2002; Paulander *et al.* 2004b).

Figure 7-4 shows an early *meta-analysis* of data from studies reporting on the association between smoking and periodontal conditions. In essence, meta-analysis is a statistical method which combines results from different studies of similar design, in order to achieve increased *power* to detect epidemiologic associations that may be difficult to identify in smaller, individual studies (Oakes 1993; Chalmers 1993; Proskin & Volpe 1994). This analysis, initially published as part of the 1996 World Workshop in Periodontics (Papapanou 1996), incorporated data from six studies, including a total of 2361 subjects with known smoking habits and periodontal status (Bergström & Eliasson 1989; Haber & Kent 1992; Locker 1992; Haber *et al.* 1993; Stoltenberg *et al.* 1993; Grossi *et al.* 1994). It can be observed that smoking was an overall increased, statistically and clinically, significant risk for severe disease (estimated overall OR of 2.82; 95% CI 2.36–3.39).

Studies examining the effects of smoking on the outcome of periodontal treatment have demonstrated that treatment responses are modified by cigarette consumption, with current smokers exhibiting poorer responses than former or never smokers (e.g. Ah *et al.*

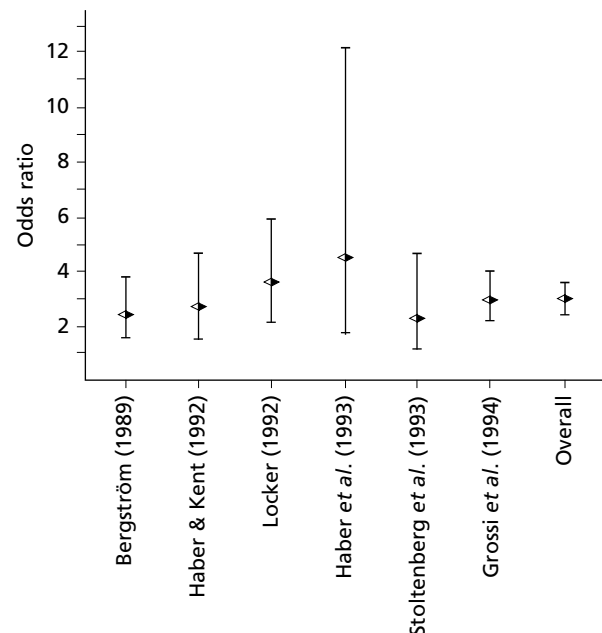


Fig. 7-4 Meta-analysis of smoking as a risk factor for periodontal disease. The studies included are: Bergström (1989), Haber & Kent (1992), Locker (1992), Haber *et al.* (1993), Stoltenberg *et al.* (1993), and Grossi *et al.* (1994). Bars indicate the 95% confidence limits for the depicted odds ratios. (Source: Papapanou 1996. Reproduced from American Academy of Periodontology.)

1994; Kaldahl *et al.* 1996; Grossi *et al.* 1997; Kinane & Radvar 1997; Tonetti *et al.* 1998; Trombelli *et al.* 2003, Stavropoulos *et al.* 2004; Paulander *et al.* 2004a; Rieder *et al.* 2004; Sculean *et al.* 2005; Wan *et al.*, 2009). Notably, these studies have confirmed the negative effect of smoking on the outcome of several periodontal treatment modalities, including non-surgical, surgical, and regenerative periodontal therapy. Meta-analyses of the effects of smoking on the outcome of periodontal therapy (Garcia 2005; Labriola *et al.* 2005; Patel *et al.* 2012) support the above conclusions.

In contrast, smoking cessation was shown to have beneficial effects on periodontal status. In a

longitudinal study (Bolin *et al.* 1993), 349 subjects with ≥ 20 remaining teeth were examined on two occasions 10 years apart (1970 and 1980). Progression of periodontal disease was assessed on radiographs at all approximal tooth surfaces and was shown to be almost twice as rapid in smokers as in non-smokers. It was also observed that subjects who quit smoking at some time point within the observation period had a significantly retarded progression of bone loss compared to that in smokers. Similar observations were made by Krall *et al.* (1997) who reported that, over a mean follow-up period of 6 years, subjects who continued to smoke had a 2.4–3.5-fold risk of tooth loss when compared to individuals who quit smoking. In a 10-year follow-up study, Bergström *et al.* (2000a) observed an increase of periodontally diseased sites concomitant with loss of periodontal bone height in current smokers, as compared to non-smokers whose periodontal health condition remained unaltered throughout the period of investigation. The periodontal health condition in former smokers was similarly stable to that of non-smokers, underscoring the beneficial effects of smoking cessation. In a shorter (12-month) follow-up study evaluating the adjunctive effect of smoking cessation on the outcome of non-surgical periodontal therapy, Rosa *et al.* (2011) showed enhanced gain in clinical attachment in chronic periodontitis patients who quit smoking when compared to their smoking counterparts. Importantly, smoking cessation alone or in conjunction with non-surgical periodontal therapy appears to result in a composition of subgingival microbiota that comprises higher levels of health-associated species and lower levels of periodontal pathogens (Fullmer *et al.* 2009; Delima *et al.* 2010).

In conclusion, cigarette smoking appears to fulfill the majority of the required steps of the risk assessment process stipulated by Beck (1994) and is considered one of the major risk factors for periodontitis.

Diabetes mellitus

The role of diabetes mellitus (DM) as a risk factor for periodontitis has been debated for decades (Genco & Löe 1993), but several biologically plausible mechanisms by which the disease may contribute to impaired periodontal conditions have been identified over the past decade [for reviews see Lalla *et al.* (2000), Mealey & Oates (2006), and Lalla and Papapanou (2011)]. Table 7-6 summarizes the epidemiologic evidence based on a number of case-control and prospective cohort studies that examined the periodontal status of patients with diabetes. A recent meta-analysis including 49 cross-sectional and eight longitudinal studies confirmed a strong association between type 2 DM and periodontitis, but concluded that the evidence for type 1 DM is weaker (Chávarry *et al.* 2009). The adverse effects of DM on periodontal status appear to be particularly pronounced in subjects with a long

duration of DM and poor metabolic control (Grossi & Genco 1998; Taylor *et al.* 1996, 1998a; Lalla *et al.* 2004). Indeed, studies have provided evidence of a dose-response relationship between poor metabolic control and the severity as well as the progression of periodontitis (Seppälä *et al.* 1993; Tervonen & Oliver 1993; Taylor *et al.* 1998a; Tervonen & Karjalainen 1997; Guzman *et al.* 2003; Bandyopadhyay *et al.* 2010). Expanding this observed dose-response relationship to the prediabetic state, the level of glucose intolerance in non-diabetic individuals was also correlated with the severity of periodontal disease (Saito *et al.* 2004). In line with the above observations, the outcome of periodontal treatment in well-controlled diabetic patients is similar to that of non-diabetic subjects (Westfelt *et al.* 1996; Christgau *et al.* 1998; Faria-Almeida *et al.* 2006), while patients with poorly controlled DM display an inferior treatment outcome (Tervonen & Karjalainen 1997).

The age of onset of DM manifestations in the periodontal tissues has been addressed in studies examining children and adolescents with type 1 DM (de Pommereau *et al.* 1992; Pinson *et al.* 1995) and both type 1 and type 2 DM (Lalla *et al.* 2006). All three studies documented more pronounced gingival inflammation in subjects with diabetes aged between 6 and 18 years. The case-control study by Lalla *et al.* (2006) further reported that clinical attachment loss was more pronounced in young patients with diabetes after adjustment for age, gender, ethnicity, gingival bleeding, and frequency of dental visits. In a subsequent publication, Lalla *et al.* (2007b) reported data on 350 children with either type 1 or type 2 DM and found a strong positive association between mean HbA1c levels over the 2 years preceding the dental examination and periodontitis. In a report including a total of 700 children, 350 with diabetes and 350 non-diabetic controls, Lalla *et al.* (2007a) documented a statistically increased periodontal destruction in children with diabetes across all disease definitions tested and in both age subgroups of 6–11 and 12–18 years.

Several studies suggest a two-way relationship between DM and periodontitis. Beyond the observed increased severity of periodontal tissue destruction in subjects with DM, studies indicate a higher incidence of DM complications and poorer metabolic control of diabetes in periodontitis patients [for review see Lalla & Papapanou (2011)]. These findings are discussed in more detail in Chapter 14.

Obesity

The biologic plausibility of a potential link between obesity and periodontitis has been suggested to involve a hyper-inflammatory state and an aberrant lipid metabolism prevalent in obesity, as well as the pathway of insulin resistance (Saito *et al.* 1998; Nishimura & Murayama 2001), all of which may collectively result in an accelerated breakdown of the

Table 7-6 Selected studies using diabetes mellitus as exposure of significance for periodontitis.

Authors/country ^a	Sample/methodology	Findings
Hugoson <i>et al.</i> (1989) Sweden	82 subjects with long and 72 with short duration IDDM; 77 non-diabetic individuals (aged 20–70 years); full-mouth, probing assessments at four sites/tooth; radiographs of lower molar–premolar regions; subjects assigned into five groups according to increasing severity of periodontal disease; no multivariate analysis	No notable difference in plaque, calculus, and number of teeth between individuals with and without DM Long duration DM patients were more frequently classified in groups 4 and 5 and had significantly more tooth surfaces with PD of ≥ 6 mm than non-diabetic controls Significantly more extensive alveolar bone loss in individuals with long duration DM aged 40–49 years old
Shlossman <i>et al.</i> (1990) USA	3219 Pima Indians, aged ≥ 5 years; prevalence of NIDDM 23% (20% in men, 25% in women); probing assessments at six sites/tooth for six index teeth; alveolar bone loss from panoramic radiographs; 2878 subjects with available radiographic data, probing assessments or both; comparison between diabetic and non-diabetic individuals with respect to AL and alveolar bone loss	Median attachment loss and alveolar bone loss higher in DM for all age groups and both genders
Emrich <i>et al.</i> (1991) USA	Sample and methodology same as above (Shlossman <i>et al.</i> 1990); 1342 Pima Indians, aged 15 years and older, with natural teeth; 19% (254) with diabetes and 12% (158) with impaired glucose tolerance; linear logistic models to predict prevalence and severity of periodontal disease; prevalence: ≥ 1 sites with AL of ≥ 5 mm or alveolar bone loss $\geq 25\%$ of the root length; severity: square root of average AL or alveolar bone loss	Diabetes, age, and calculus were significant risk markers for periodontitis OR for an individual with DM to have PD was 2.8 (clinically assessed) and 3.4 (radiographically)
de Pommereau <i>et al.</i> (1992) France	85 adolescents with IDDM, aged 12–18 years, and 38 healthy age-matched controls; probing assessments at six sites/tooth, all teeth; bite-wing radiographs at molars and sites with AL > 2 mm; patients divided according to disease duration ($>$ or < 6 years); sexual maturation according to Tanner's classification; metabolic control expressed through glycosylated hemoglobin (HbA1c); non-parametric pair-wise analysis	None of the subjects had sites with AL ≥ 3 mm or radiographic signs of periodontitis Despite similar plaque scores, diabetic children had significantly more gingival inflammation No significant relation between gingival condition and age, Tanner's index, HbA1c level or disease duration
Oliver & Tervonen (1993) USA	114 diabetic patients, aged 20–64 years (60% with IDDM and 40% with NIDDM); half-mouth, probing assessments at four sites/tooth; data from the 1985–86 National Survey served as controls	Tooth loss was similar among diabetic and non-diabetic US employed adults 60% of individuals with DM and 16% of controls had ≥ 1 site with PD ≥ 4 mm AL data were comparable in both groups
Thorstensson & Hugoson (1993) Sweden	83 IDDM patients and 99 age- and sex-matched non-diabetic controls, aged 40–70 years; full-mouth, probing assessments at four sites/tooth; radiographs of lower molar–premolar regions; subjects assigned to five groups according to increasing severity of periodontal disease; univariate analysis	DM patients aged 40–49 years (mean disease duration 25.6 years) had more periodontal pockets ≥ 6 mm and more extensive alveolar bone loss than non-diabetic controls, but this was not the case for subjects aged 50–59 or 60–69 years (mean disease duration 20.5 and 18.6 years, respectively) Disease duration appeared to be a significant determinant of development of periodontitis
Pinson <i>et al.</i> (1995) USA	26 IDDM children, aged 7–18 years, and 24 controls, 20 of whom were siblings of the diabetic patients; full-mouth, probing assessments at six sites/tooth; metabolic control assessed through HbA1c; analysis of co-variance	Overall, no statistically significant differences between cases and controls No association between levels of HbA1c and clinical variables After correcting for plaque, diabetics showed more severe gingival inflammation in specific tooth regions
Bridges <i>et al.</i> (1996) USA	A samples of 233 men, aged 24–78 years, including 118 diabetic (46 type 1 and 72 type 2) and 115 non-diabetic subjects, matched for age and BMI	Plaque and gingivitis, bleeding scores, PD, AL and missing teeth were significantly higher in diabetic than non-diabetic men

(Continued)

Table 7-6 Continued

Authors/country ^a	Sample/methodology	Findings
Tervonen & Karjalainen (1997) Finland (L)	36 patients with type 1 diabetes and 10 controls, aged 24–36 years, received non-surgical periodontal therapy and were followed at 4 weeks, 6 and 12 months; patient with diabetes were further grouped according to diabetic status as: D1 (n = 13) with no diabetic complications and good long-term metabolic control; D2 (n = 15) moderate metabolic control with/without retinopathy; D3 (n = 8) severe diabetes with poor metabolic control and/or multiple complications; periodontal status was monitored radiographically	The periodontal status of the diabetic patients with good control and no complications (D1) and those with moderate control (D2) was similar to non-diabetic controls Diabetic subjects with poor metabolic control and/or multiple complications (D3) exhibited higher extent of AL ≥ 2 mm at baseline and higher recurrence of PD ≥ 4 mm during follow up
Taylor <i>et al.</i> (1998a) USA (L)	2-year study of 21 patients with type 2 diabetes, including 14 with poor and 7 with better metabolic control, and 338 controls, aged 15–57 years, Native Americans; progression of bone loss was assessed on radiographs; co-variables included age, calculus, gingival and plaque indices, time to follow-up, alcohol consumption, smoking, obesity (BMI >27), coronary heart disease, and gender	In multiple logistic regression, poorly controlled diabetic subjects were 11 times more likely (95% CI 2.5–53.3) to have more pronounced bone loss progression than non-diabetic subjects; no such differences were found between better controlled and non-diabetic controls Age, time to follow-up, pronounced bone loss at baseline, and calculus index were significant predictors of bone loss progression.
Taylor <i>et al.</i> (1998b) USA (L)	2-year study of 24 subjects with NIDDM and 362 subjects without diabetes, aged 15–57 years; degree of bone loss on panoramic radiographs was assessed on a scale of 0–4	A regression model having progression of bone loss as the dependent variable revealed a cumulative OR for NIDDM of 4.23 (95% CI 1.8–9.9); the association was modified by age, with younger adults exhibiting higher risk for alveolar bone loss progression
Lalla <i>et al.</i> (2007) USA	Case–control study of 350 children and adolescents, aged 6–18 years, with diabetes (325 with type 1 diabetes; 25 with type 2 diabetes) and 350 non-diabetic controls; half-mouth examination at four sites/tooth in all fully erupted teeth with respect to PD and AL; three different definitions of periodontitis were used	Children with diabetes had significantly more plaque and gingival inflammation than controls and a higher number of teeth with AL Regression analysis controlling for age, gender, ethnicity, gingival bleeding, and frequency of dental visits revealed that diabetes was statistically related to periodontal destruction across all disease definitions (OR 1.84–3.72)

^aL indicates a longitudinal study.

PD, probing depth; AL, attachment level; ; BMI, body mass index; OR, odds ratio; CI, confidence interval.

DM: diabetes mellitus.

IDDM and NIDDM, insulin-dependent and non-insulin dependent diabetes mellitus, respectively; these diagnostic terms are no longer used and have been replaced by type 1 and type 2 diabetes.

periodontal tissues. Indeed, a number of studies have indicated a positive association between obesity, defined as body mass index (BMI) ≥ 30 kg/m², and periodontitis. Four publications have documented such an association in the NHANES III database. Wood *et al.* (2003), using a subset including Caucasian subjects aged 18 years and older, reported that BMI, waist-to-hip ratio, visceral fat, and fat-free mass were associated with periodontitis after adjusting for age, sex, history of diabetes, current smoking, and socio-economic status. Al-Zahrani *et al.* (2003) reported a significant association between both BMI and waist-to-hip ratio and periodontitis in younger adults, but no association in middle-aged or older adults. Genco *et al.* (2005) reported that overweight subjects in the upper quartile of the insulin resistance index were 1.5 times more likely to have periodontitis compared to their counterparts with a high BMI but a low insulin

resistance index. Finally, Andriankaja *et al.* (2010) demonstrated an association between *metabolic syndrome* (i.e. a combination of hypertension, impaired fasting glucose, large waist circumference, and dyslipidemia) and periodontitis in women, and between abdominal obesity and periodontitis in both women and men.

In a longitudinal study of 1038 healthy, Caucasian US male veterans, obesity conferred a 41–72% increased risk for progression of periodontitis, after adjustment for several co-variables (Gorman *et al.* 2012).

Corroborating data have been reported also from countries other than the US. In a sample of 643 apparently healthy Japanese adults, Saito *et al.* (2001) reported that waist-to-hip ratio, BMI, and body fat were significant risk indicators for periodontitis after adjustments for known risk factors. In a recent longitudinal study from Japan involving a sample of 3590

individuals, the 5-year incidence of periodontitis was statistically higher for both those with a BMI between 25 and 30 kg/m² and those with a BMI of ≥ 30 kg/m², when compared to individuals with a BMI of ≤ 22 kg/m² (Morita *et al.* 2011), establishing a dose-response relationship between overweight/obesity and risk for periodontitis. Finally, in a study involving a nationally representative sample of 7188 subjects in Korea, *metabolic syndrome* was associated with periodontitis (Kwon *et al.* 2011). In contrast, an inverse association between obesity and clinical attachment loss was observed in a study involving 1579 men and women in Denmark (Kongstad *et al.* 2009).

Given that (1) the majority of the above publications are cross-sectional, and thus do not facilitate inferences on temporality or mechanisms, and (2) the available epidemiologic data are limited and not universally consistent, additional research on the role of obesity in periodontitis is warranted.

Osteopenia/osteoporosis

Several early cross-sectional studies, of limited sample size and largely confined to post-menopausal women, have suggested that women with low bone mineral density are more likely to have gingival recession and/or pronounced gingival inflammation and clinical attachment loss (von Wöhrn *et al.* 1994; Mohammad *et al.* 1996, 1997; Tezal *et al.* 2000). In a radiographic study of 1084 subjects aged 60–75 years, Persson *et al.* (2002) reported a positive association between osteoporosis and periodontitis with an OR of 1.8 (95% CI 1.2–2.5). However, studies that have failed to report such an association have also been published (Weyant *et al.* 1999; Lundström *et al.* 2001).

Based on these observations, it has been hypothesized that the systemic loss of bone density in osteoporosis may, in combination with hormone action, heredity, and other host factors, provide a host system that is increasingly susceptible to inflammation-associated destruction of the periodontal tissues (Wactawski-Wende 2001). In a cross-sectional study of 1329 post-menopausal women in the US, systemic bone density was positively associated with clinical attachment loss in women with subgingival calculus, but negatively associated in women without calculus (Brennan *et al.* 2007). The data from longitudinal studies are apparently conflicting. Payne *et al.* (1999, 2000) reported an enhanced longitudinal alveolar bone loss in osteoporotic women versus women with normal mineral bone density, Yoshihara *et al.* (2004) found, after adjustments, a significant association between bone mineral density and 3-year longitudinal attachment loss in Japanese subjects aged ≥ 70 years. In contrast, Reinhardt *et al.* (1999) reported no significant impact of serum estradiol levels on longitudinal attachment loss over a 2-year period. Recent systematic reviews of the available studies on osteoporosis and periodontitis (Martinez-Maestre *et al.* 2010; Megson *et al.* 2010) concluded that the relationship

between the two conditions remains unclear. Additional prospective studies are warranted to clarify the inter-relationship between the two diseases and its clinical implications.

Human immunodeficiency virus infection

Studies published in the late 1980s seemed to indicate that both the prevalence and the severity of periodontitis were exceptionally high in patients with acquired immunodeficiency syndrome (AIDS) (Winkler & Murray 1987), but a more tempered picture emerged in subsequent publications. While it cannot be ruled out that the initial reports included biased samples, it is also possible that the successful control of immunosuppression in human immunodeficiency virus (HIV)-positive subjects by means of high activity antiretroviral therapy (HAART) and other continuously evolving drugs has influenced the incidence of periodontal disease progression in HIV-seropositive subjects and has resulted in less severe periodontal manifestations of HIV infection (Chapple & Hamburger 2000). For example, a cross-sectional study of 326 HIV-infected adults (McKaig *et al.* 1998) revealed that, after adjustments for CD4 counts, persons taking HIV antiretroviral medication were five times less likely to suffer from periodontitis than those not taking such medication, underscoring the importance of the host's immunologic competency in this context.

Nevertheless, publications continue to generate conflicting results. Thus, although a number of studies (Smith *et al.* 1995a; Robinson *et al.* 1996; Ndiaye *et al.* 1997; McKaig *et al.* 1998; Nittayananta *et al.* 2010; Stojkovic *et al.* 2011) have indicated higher prevalence and severity of periodontitis in HIV-positive subjects when compared to controls, other studies have either not supported this notion or have indicated that the differences in periodontal status between HIV-seropositive and -seronegative subjects are limited (Cross & Smith 1995; Lamster *et al.* 1997; Scheutz *et al.* 1997; Lamster *et al.* 1998; Vastardis *et al.* 2003). Studies investigating the pathobiology of periodontitis in HIV-infected subjects suggested that specific IgG subclass responses to periodontopathic bacteria were similar in HIV-positive and HIV-negative subjects (Yeung *et al.* 2002), while CD4 count levels were not found to correlate with the severity of periodontitis (Martinez Canut *et al.* 1996; Vastardis *et al.* 2003).

The few available longitudinal studies are equally conflicting. Two companion publications, from a short-term follow-up study (Smith *et al.* 1995b; Cross & Smith 1995) involving a group of 29 HIV-seropositive subjects who were examined at baseline and at 3 months, reported a low prevalence and incidence of clinical attachment loss. The subgingival microbial profiles of the seropositive subjects resembled those of non-systemically affected subjects, and did not correlate with their CD4 and CD8 lymphocyte counts. Similarly, in a small follow-up study of 12 months' duration, Robinson *et al.* (2000) found no difference in the

progression of periodontitis between HIV-positive and HIV-negative subjects. Hofer *et al.* (2002) demonstrated that compliant HIV-positive subjects can be successfully maintained in a manner similar to non-infected controls. However, a 20-month follow-up study of 114 homosexual or bisexual men (Barr *et al.* 1992) revealed a clear relationship between incidence of clinical attachment loss and immunosuppression, expressed through CD4 cell counts. The authors suggested that seropositivity in combination with older age confers an increased risk for attachment loss. Similar observations were reported by Lamster *et al.* (1997), who concluded that periodontitis in the presence of HIV infection is dependent upon the immunologic competency of the host as well as the local inflammatory response to the subgingival microbiota.

It appears therefore that there is no consensus in the literature on the association of HIV/AIDS and periodontitis. Variation in the severity of oral pathologic conditions due to ongoing advancements in HIV/AIDS therapy will likely further contribute to the diversity of the findings (Freed *et al.* 2005).

Psychosocial factors

The mechanisms by which psychosocial stress may affect the periodontal status are complex. It has been suggested that one of the plausible pathways may involve behavioral changes leading to smoking and poor oral hygiene that, in turn, may affect periodontal health (Genco *et al.* 1998). In the absence of an unequivocal biologic measure of stress, a limited number of studies have used proxy measures of stress to study its association with periodontitis. In a study of 1426 subjects in Erie County, NY, USA, Genco *et al.* (1999) reported that adults who were under financial strain and exhibited poor coping behaviors were at increased risk of severe periodontitis when compared with subjects who demonstrated good coping behavior patterns under similar financial strain, or with controls under no financial strain. In a sample of 1089 adults in rural Japan, job- and health-related stress was positively associated with clinical attachment loss after adjustments for common risk factors (Akhter *et al.* 2005). War-related stress was found to be associated with poor periodontal conditions in Croatia (Spalj *et al.* 2008). Similar observations were made in a study of an immigrant population from Ethiopia, in which psychological distress was positively associated with deep periodontal pockets (Vered *et al.* 2011). In contrast, a study of 681 subjects carried out in Lithuania (Aleksiejuniene *et al.* 2002) could not document an association between psychosocial stress and periodontitis, although it reported that the disease did correlate with lifestyle factors. In a small prospective study based on 23 employed adults, Linden *et al.* (1996) reported that longitudinal attachment loss was significantly predicted by increasing age, lower socioeconomic status, lower job satisfaction, and type

A personality, characterized by aggressive, impatient, and irritable behavior.

Clearly, the role of stress in periodontitis has not been fully explored and multiple gaps in our knowledge exist. Nevertheless, given the established role of the sympathetic, parasympathetic, and peptidergic/sensory nervous systems, as well as that of the hypothalamic–pituitary–adrenal axis on brain-to-immune regulatory pathways, such a role is clearly biologically plausible. Experimental animal studies have begun to shed light on basic mechanisms that may explain the link between psychosocial factors and periodontitis. For example, a study by Breivik *et al.* (2006) demonstrated that experimentally-induced depression accelerated tissue breakdown in a ligature-induced periodontitis rat model and that pharmacologic treatment of depression attenuated this breakdown. In a study in humans, salivary cortisol levels (indicative of psychological stress) were positively associated with the extent and severity of periodontitis (Hilgert *et al.* 2006).

Additional basic and epidemiologic research is needed to fully elucidate a possible relationship between psychological factors and periodontal disease.

Concluding remarks

The analytical epidemiologic studies described in this chapter are obviously diverse with respect to important elements of design and methodology, such as definitions of disease, sample size, use of full-mouth or partial-mouth recording protocols, length of follow-up in longitudinal studies, adjustment for potential confounders, etc. Nevertheless, despite these apparent shortcomings, a number of conclusions can be made with reasonable certainty:

1. Specific bacteria, cigarette smoking, and diabetes mellitus are the major established risk factors for periodontitis. A number of biologically plausible, potentially important additional factors are in need of further investigation in future studies.
2. There is a need to introduce uniform definitions of periodontitis to be used in analytical epidemiologic studies. This will facilitate valid comparisons, establish whether seemingly conflicting data reflect true biologic variation or are exclusively owed to methodologic inconsistencies, and contribute to the correct identification of risk factors. The definitions proposed by the Consensus Report of the 5th European Workshop in Periodontology (Tonetti & Claffey 2005), the CDC/APP definition (Page & Eke 2007), and the definition for *aggressive periodontitis* in epidemiologic studies (Demmer & Papapanou 2010) may serve such a purpose. Obviously, no definition is devoid of shortcomings and the above proposals are no exception. In the absence of a universally accepted consensus definition, data presentation using several of the above definitions should be encouraged.

3. Studies need to distinguish clearly between risk factors and disease predictors. Although the use of the latter as explanatory variables in multivariate models may increase the coefficient of determination (i.e. the proportion of the variance explained by means of the models), it may also obscure the significance of true etiologic factors. For example, as shown by Ismail *et al.* (1990), factors with biologically plausible etiologic potential (such as dental plaque) may not retain their significance in multivariate models that include alternative expressions of disease such as tooth mobility. It has been demonstrated that baseline levels of disease and morphologic features such as angular bony defects are powerful predictors of future disease progression (Papapanou *et al.* 1989; Papapanou & Wennström 1991). Haffajee *et al.* (1991a) showed that age, plaque, and bleeding on probing are related to baseline disease levels as well as to incident disease. In the search for true exposures of significance for disease onset or progression, inclusion of a factor in a model may thus erroneously discredit another co-varying, biologically significant factor.

An interesting observation was made in a longitudinal study by Beck *et al.* (1995), in which characteristics of patients experiencing clinical attachment loss at previously non-diseased sites were compared with those of patients suffering progression of already established disease. While low income and medication with drugs associated with soft tissue reactions were features common to both groups of patients, new sites with attachment loss were more frequent in patients who used smokeless tobacco and had a history of oral pain. Risk for progression of established disease was higher in cigarette smokers, subjects with high levels of subgingival *P. gingivalis*, and individuals with worsening financial problems. These data suggest that periodontitis may be like other diseases for which the factors associated with the initiation of the disease may be different from the ones involved in its progression. Such a distinction between factors associated with initiation and with progression may have implications for future assessment strategies and may improve the accuracy of the risk/prediction models.

One of the issues related to the descriptive epidemiology of periodontal infections that is still under debate is whether their worldwide prevalence has been decreasing over the last decades. Unfortunately, the data do not allow a clear answer for a number of reasons. First, no universal conclusion is possible, since the prevalence of periodontal disease appears to vary with race and geographic region. Second, the quality of the data available from developing and developed countries is clearly not comparable. While some well-conducted epidemiologic surveys that provide detailed information have been carried out in a number of countries, the majority of the studies in the

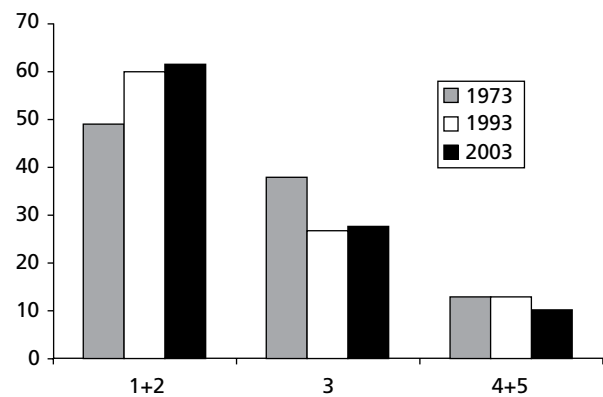


Fig. 7-5 Frequency distribution of subjects with healthy periodontal conditions or gingivitis (groups 1 + 2), moderate periodontitis (group 3), and advanced and severe periodontal disease (groups 4 + 5), in a Swedish cohort in 1973, 1993, and 2003. For definitions see text. (Adapted after personal communication with Anders Hugoson, based on data from Hugoson *et al.* 1992, 1998, 2006).

developing world have used the CPITN system, which produced data of inadequate detail. Moreover, studies using the exact same methodology to evaluate random samples drawn from the same population over time are sparse. Among the few exceptions, data available from some parts of the world, notably the US, are suggestive of such a trend for decreasing prevalence of periodontitis (Dye *et al.* 2007). A series of studies from Sweden (Hugoson *et al.* 1992, 1998b, 2005) that documented, by clinical and radiographic means, the frequency distribution of various levels of severity of periodontitis in four cross-sectional studies over a 30-year period (in 1973, 1983, 1993, and 2003). In these studies, subjects were grouped according to the severity of their periodontal conditions into five groups: groups 1 and 2 included subjects who were periodontally healthy or only had gingivitis; group 3 included subjects with moderate periodontitis, that is whose loss of periodontal tissues support did not extend beyond one-third of the root length; and groups 4 and 5 included subjects with more severe destructive disease. As shown in Fig. 7-5, a clear increase in the frequency of subjects in groups 1 and 2 was noted over the 30-year period, from 49% in 1973 to 60% in 1993 to almost 62% in 2003. This increase occurred primarily at the expense of group 3, which declined from 38% in 1973 to 27% in 1993 and apparently reached a plateau at 28% in 2003. Nevertheless, the frequency of subjects in groups 4 and 5 was virtually stable over the 30-year period: 13% in 1973, 13% in 1993, and 10.5% in 2003. Based on these data derived from a population with arguably the best access to and utilization of oral health care in the world, it may be concluded that the fraction of the population which is most susceptible to severe periodontitis is apparently not declining in frequency. Instead, the main beneficiaries of the improved oral health awareness, access to care, and increased utilization of therapeutic resources that has occurred over the last decades are likely the individuals with

moderate levels of periodontitis whose prevalence is clearly lower.

It has also been well-documented in these and other studies that the rate of edentulism has decreased substantially over the past 30 years, with elderly groups retaining their natural dentition and higher mean numbers of teeth than their counterparts a generation ago (Kassebaum *et al.* 2014). This fact *per se* should contribute to an increased prevalence of periodontal disease in older age cohorts, since retained teeth in the elderly are more likely to experience substantial cumulative attachment loss which forms the basis of the assessment of prevalence (Douglass & Fox 1993). It has been argued, however, that such a potential increase may not necessarily result in increased need for periodontal therapy (Oliver *et al.* 1989). Additional research is clearly required to further elucidate these issues, and an adequate and consistent epidemiologic methodology is essential for generating valid comparative data. Arguably, the principle task of future epidemiologic

research is to elucidate the determinants of susceptibility to severe periodontitis, prior to the development of irreversible tissue damage (Papapanou 2012, 2014). Although several risk factors have been established and a wide array of disease markers has been recognized, the impact of intervention addressing such factors on the state of periodontal health on a population level has yet to be documented. To assess the magnitude of the clinical benefit achieved by such modulation, prospective, long-term epidemiologic surveys have to be conducted.

Acknowledgments

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Part 3: Microbiology

- 8** Dental Biofilms, 169
Philip David Marsh
- 9** Dental Calculus, 183
Dieter D. Bosshardt and Niklaus P. Lang
- 10** Periodontal Infections, 191
Michael Curtis
- 11** Peri-implant Infections, 222
Lisa Heitz-Mayfield, Ricardo P. Teles, and Niklaus P. Lang

